Optimizing Maintenance: Evaluating Predictive Accuracy in ML Models for Early Fault Detection



Author: Prajwal Gurivindapalli

Matriculation: 30630894

Degree: Master of Science (M.Sc.) in Data Science

First Supervisor: Prof. Dr. Iftikhar Ahmed

Second Supervisor: Shan Faiz

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University of Europe for Applied Sciences, Potsdam

Author Declaration

I hereby declare that this thesis is my own work and that I have clearly acknowledged all sources used in its creation. No portion of the work referred to in this thesis has been submitted in support of an application for another degree or qualification at this or any other university or institute of learning.

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Abstract

Industry 4.0 technologies have revolutionized the way manufacturing is done today by allowing manufacturers to monitor and make decisions with the help of smart technologies. One of these developments, predictive maintenance, is proven to be a valuable tool to reduce unexpected equipment downtime, maintenance costs and boost operational efficiency. Unlike the traditional reactive and preventive maintenance approaches, predictive maintenance leverages machine learning algorithms that can analyze data from various sensors to anticipate potential failures before they happen.

This research aims to explore supervised machine learning methods to support predictive maintenance with the AI4I Predictive Maintenance Dataset. The main goal is to train and evaluate 3 machine failure classification algorithms: Logistic Regression, Random Forest and Extreme Gradient Boosting (XGBoost) using operational sensor data. The effect of critical process parameters such as air temperature, process temperature, rotational speed, torque, tool wear and machine type on machine failure prediction is also discussed.

A systematic machine learning pipeline was employed that involved data preprocessing, EDA, feature engineering, class balancing, model building, and model testing. Standard classification metrics like accuracy, precision, recall, F1-score and confusion matrices were used to evaluate the models. Additionally, feature importance analysis was conducted to show which operational variables impact the prediction most.

The outcomes of the experiments prove that machine learning algorithms can be used to predict machine failures with industrial sensor data. Ensemble learning methods outperform traditional linear classification methods, with the highest predictive performance being from XGBoost followed by Random Forest and Logistic Regression. Moreover, tool wear, torque and rotational speed were found as the most important parameters affecting the prediction of machine failures using the feature importance analysis technique.

This study's results highlight the feasibility of developing predictive maintenance systems using machine learning for proactive maintenance practices, minimizing un-

expected maintenance downtimes, and enhancing equipment reliability. Additionally, the study offers a comparative analysis of well-established machine learning algorithms on a public industrial dataset, giving insight for future studies and industrial applications.

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Chapter 1 Introduction

1.1 Background of the Study

Digital technologies have revolutionised the industrial sector, with the result that Industry 4.0 has emerged as a paradigm Lasi et al. (2014), Xu et al. (2018). It brings together the field of intelligent systems, automation, advanced data analytics, and the Industrial Internet of Things (IIoT) to achieve operational efficiency Lasi et al. (2014), Zheng et al. (2018). Today's factories make extensive use of sensors and monitoring devices that are interconnected in order to keep tabs on the condition of machines in real-time Lee, Bagheri & Kao (2015), Tao et al. (2018). Therefore, enterprises are changing the way they maintain their systems to adopt data-driven operational strategies.

In order to ensure the continuous production of industries, the reliability and availability of equipment are very important. Unauthorized outages result in substantial outages, financial losses, reduced product quality, and workplace safety risks Mobley (2002). There were two key paradigms in the past for industrial operations: reactive maintenance (reacting after a failure has occurred) and preventive maintenance (maintaining equipment according to a fixed schedule) Mobley (2002), Jardine et al. (2006). But both methods have their own drawbacks, including high labor costs, unexpected failures during operation, or premature replacement of components Jardine et al. (2006), Si et al. (2011).

Predictive maintenance (PdM) is now being offered as a viable solution to overcome these limitations. Predictive maintenance takes a proactive approach to maintenance by using real-time sensor streams and historical performance logs, predicting the degradation of equipment before failure can occur Matzka (2020), Jardine et al. (2006). With the increasing availability of high-frequency operational data, the concept of predictive maintenance has gained popularity in many industries including manufacturing, aerospace, automotive, energy, and healthcare Ran et al. (2019).

Machine learning (ML) is one of the fundamental enablers for predictive maintenance applications Ran et al. (2019), Mitchell (1997). The goal of ML algorithms is to process large amounts of multidimensional sensor data, detect complex patterns,

and model non-linear failure mechanisms Hastie et al. (2009), Bishop (2006). This allows for a shift in approach from reactive operational decisions to proactive asset management decisions Ran et al. (2019), Mobley (2002).

In this study, supervised machine learning algorithms namely Logistic Regression, Random Forest, and XGBoost are evaluated for predicting equipment failures based on measurements taken from industrial sensors Breiman (2001), Chen & Guestrin (2016), Hosmer Jr et al. (2013). In addition, this study identifies the most important operational parameters associated with machinery failures to guide predictive maintenance strategies Matzka (2020), Lundberg & Lee (2017).

Maintenance management undergoes a significant transformation when Industry 4.0 frameworks take hold Lasi et al. (2014), Xu et al. (2018). Lee et al. Lee, Bagheri & Kao (2015) emphasized the importance of continuously feeding cloud-based systems with machine health information for performing real-time machine diagnosis. Subsequent empirical synthesis work by Jardine et al. Jardine et al. (2006) and Ran et al. Ran et al. (2019) showed that machine learning approaches to maintenance can substantially lower unexpected downtime, operational costs, and increase asset life compared to traditional maintenance approaches.

Modern manufacturing systems do not rely on static schedules but continuously acquire sensor data for key operational metrics like process temperature, rotational speed, torque, vibration, and tool wear Matzka (2020), Xu et al. (2018). These data streams feed into intelligent models capable of detecting complex failure mechanisms Lei et al. (2018), Zheng et al. (2017). Surveying the current state of the art, Ran et al. Ran et al. (2019) showed that data-driven predictive maintenance is one of the most vital application areas of AI in modern smart manufacturing.

1.2 Industry 4.0 and Predictive Maintenance

Cyber-physical systems, cloud computing, artificial intelligence, big data analytics, and Internet of Things (IoT) connectivity are the pillars of Industry 4.0, according to conceptual definitions of Industry 4.0 by Lasi et al. Lasi et al. (2014) and Xu et al. Xu et al. (2018). These technologies enable continuous communication between equipment and central systems, resulting in inter-connectivity across factory environments Lee, Bagheri & Kao (2015), Zheng et al. (2018). One of the main goals of Industry 4.0 is optimizing operational productivity with minimized resource usage and costs Lasi et al. (2014). Data analytics and predictive maintenance are well suited for this goal: by leveraging data and analytical tools, predictive maintenance can help optimize servicing operations and maximize asset uptime Jardine et al. (2006), Ran et al. (2019).

The operational advantages of predictive maintenance include lower repair costs, longer asset life cycles, better work safety, and reduced downtime Mobley (2002), Si et al. (2011). A central challenge in this field is managing multiple streams of sensor data and recognizing complex interactions between features that lie beyond the capabilities of traditional statistical models, a task where machine learning algorithms are crucial Hastie et al. (2009), Pedregosa et al. (2011).

With Industry 4.0, the shift towards intelligent, data-driven maintenance strategies has gained momentum Lasi et al. (2014). Lee et al. Lee, Ardakani, Yang & Bagheri (2015) showed how to use IIoT sensor networks and cloud computing infrastructures to deliver real-time operational metrics for predictive machine health monitoring.

Predictive maintenance differs from preventive maintenance because it uses real-time data from machine sensors to assess equipment condition instead of setting fixed schedules. The theoretical framework for machinery diagnostics and condition-based monitoring was established by the early work of Jardine et al. Jardine et al. (2006). In more recent machine learning applications, studies demonstrate that algorithms trained on real operating data can considerably decrease redundant maintenance actions and maintenance costs Matzka (2020), Ran et al. (2019).

Supervised machine learning algorithms are effective for modeling complex interactions between operational variables, such as rotational speed, torque, process temperature, air temperature, and tool wear Matzka (2020). Random Forest was introduced by Breiman Breiman (2001) and has proven effective in ensembling decision trees while resisting overfitting. XGBoost, introduced by Chen and Guestrin Chen & Guestrin (2016), is an optimized gradient-boosting framework designed for efficient tabular data modeling. For complex industrial datasets, ensemble methods (Random Forest and XGBoost) are generally superior in predicting industrial failures compared to traditional linear models Breiman (2001), Chen & Guestrin (2016), Caruana & Niculescu-Mizil (2006).

1.3 Problem Statement

Failure of industrial equipment is still a significant problem in industrial manufacturing industry all over the world. Equipment Breakdowns lead to loss of production time, increased repair expenses, missed production deadlines, and increased safety risks. In many cases traditional maintenance methods do not achieve the optimal balance between reliability and cost-effectiveness:

Reactive Maintenance: Maintenance efforts are done when the machine fails. This will reduce the initial monitoring expenses, but result in substantial unplanned

downtime, secondary structural damage and expensive emergency repairs.

Preventive Maintenance: Servicing at scheduled time or usage levels. This decreases unnecessary downtime but often results in unnecessary maintenance of fully operational pieces of equipment, resulting in unnecessary labor expenses and downtime.

The use of industrial sensor streams for continuous monitoring provides a much more efficient alternative, known as predictive maintenance. Machine learning algorithms are capable of learning complex patterns of operations and identifying early warning signs of degradation in a variety of measurements including temperature, tool wear, and rotational speed well before they are likely to cause catastrophic failure.

But the question of choosing the best machine learning architectures and determining the most direct causes of equipment failure in this regard is still a big engineering challenge. Although new predictive methods have been developed, many industrial plants have not yet been able to switch to the new maintenance methods because of integration complexity, preferring to follow the traditional corrective or periodic maintenance approach. Early studies were based on typical statistical techniques that showed to be inadequate to model the non-linear distribution of data, and many of the modern studies were based on proprietary industrial data making it difficult to validate across domains or evaluate openly. Therefore, comprehensive comparative assessments with available benchmark industrial data sets are required to systematically compare the accuracy, speed, and interpretability of features and models for real-world use.

1.4 Research Aim

The goal of this research is to create and compare supervised machine learning models for predictive maintenance and early fault detection from industrial sensor data. The study systematically tests the predictive accuracy of various machine learning algorithms for the classification of machine failure and investigates the weight of various operational variables in the machine failures.

1.5 Research Objectives

The following objectives have been set to achieve the overall research goal:

Data Collection and Exploratory Analysis: Collecting and analyzing multiple sensor industrial data systematically to gain insight into the operational failure mode, signal anomaly detection and characterization of machine degradation pat-

terns.

Data Preprocessing and Imbalance Mitigation: Raw sensor streams were cleaned, normalized, and preprocessed for supervised machine learning development, and a method to address severe class imbalance between normal operations and rare failure events was employed, known as the Synthetic Minority Over-sampling Technique (SMOTE).

Model Implementation and Pipeline Development: Building and training a variety of supervised learning models from a simple linear model (Logistic Regression) to ensemble models (Random Forest and XGBoost) for predicting impending equipment failures.

Ensure Robust Validation Framework: To ensure a strong validation framework, the implemented models are evaluated and compared using standard evaluation metrics such as Precision, Recall, F1-Score, Matthews Correlation Coefficient (MCC).

Feature Importance and Interpretability Analysis: To use the feature importance analysis methods to determine the most important sensor parameters and operational variables affecting the ability to predict machinery failures.

Industrial Deployment Framework: To transform empirical results into real-world practical recommendations and architectural advice on how to integrate predictive maintenance systems into Industry 4.0 and smart manufacturing.

1.6 Research Questions

The following research questions are presented in relation to the research aim and research objectives:

RQ1: How accurate are predictions on failure of industrial machinery based on supervised machine learning models using sensor data? This item assesses the ability of machine learning algorithms to identify potential machine failures based on the various operational metrics.

RQ3: What is the best machine learning algorithm to predict the performance in industrial maintenance applications? The goal of this question is to compare the prediction strength of Logistic Regression, Random Forest, and Extreme Gradient Boosting (XGBoost) to determine which is best.

RQ3: What are the most critical operating conditions to be used to predict machine failures? The key operational variables (air temperature, process temperature, rotational speed, torque, tool wear, machine type) are correlated and selected via a feature selection analysis to evaluate their relative importance.

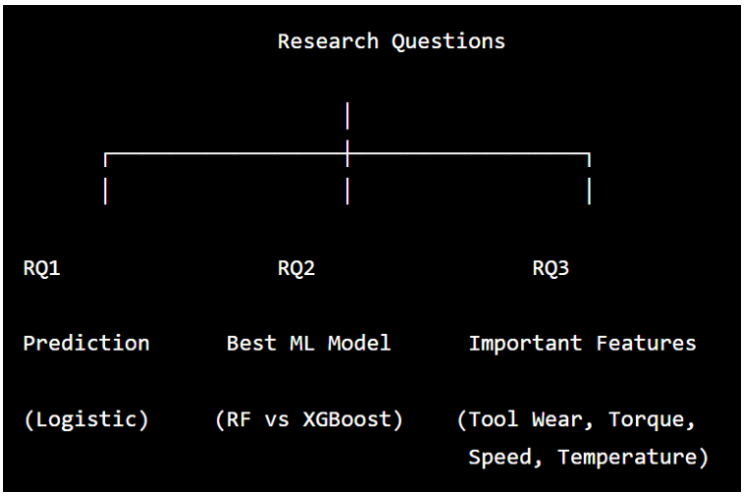


Figure 1.1: Structured hierarchy of the research questions (RQ1–RQ3)

1.7 Contributions and Impact of the Research

This research contributes significant academic insights, practical operational guidance, and technical frameworks to modern industrial engineering and data-driven reliability management.

From an academic perspective, this work addresses a critical gap in the literature by providing a systematic, empirical comparison of diverse supervised machine learning architectures ranging from traditional linear models to complex ensemble algorithms operating under real-world data constraints. By evaluating Logistic Regression, Random Forest, and XGBoost within a strictly standardized environment, this study delivers concrete evidence regarding algorithmic trade-offs, predictive robustness, and sensitivity to severe class imbalance. Furthermore, the integration of feature importance analysis enhances model interpretability, offering a theoretical baseline for understanding how machine learning models internalize physical equipment degradation patterns.

From an operational perspective, the findings offer actionable decision-support tools for plant managers, maintenance engineers, and reliability specialists seeking to transition from traditional preventative maintenance to proactive, predictive regimes. By identifying the specific operational parameters and sensor channels that contribute most significantly to machinery failure, this study enables organizations to streamline sensor networks, optimize condition-monitoring schedules, and reduce hardware redundancy. Ultimately, these practical insights help target maintenance interventions precisely when needed, minimizing catastrophic failure risks while avoiding unnecessary servicing costs.

From a technical and computational perspective, this thesis establishes a fully re-

producible end-to-end data pipeline tailored to high-dimensional sensor streams. By demonstrating how advanced data preprocessing, synthetic oversampling, and model evaluation techniques come together to solve complex industrial classification problems, this research provides a practical implementation blueprint. As smart manufacturing environments continue to evolve, the methodologies established in this work offer a scalable foundation for deploying scalable predictive analytics within Industry 4.0 infrastructure.

1.8 Research Boundaries and Constraints

The research is centered on predictive maintenance systems that rely on data and machine learning models on data from industrial sensors. The public AI4I Predictive Maintenance Dataset is used in the investigation, consisting of process temperature, air temperature, rotational speed, torque, tool wear and classifications of machine failure.

The theoretical and technical area of interest includes supervised machine learning models such as Logistic Regression, Random Forest and XGBoost. The machine failure classification models are tested for binary classification for whether a machine will fail or not given the operational inputs. The accuracy, precision, recall, F1-score and ROC-AUC are used to evaluate the model performance.

Conclusions provide insight into predictive maintenance design, but results are analyzed in the context of the operational environment of the target benchmark data set and may need validation for use in other manufacturing settings when applied to new environments.

1.9 Structure of the Thesis

This thesis is structured into six distinct chapters:

- Chapter 1 (Introduction): Presents the general background of the study, problem statement, research aim, objectives, research questions, significance of the study, and the overall structural framework.
- Chapter 2 (Background): Summarizes current theoretical and empirical research related to Industry 4.0, Condition-Based Monitoring, Predictive Maintenance, and Machine Learning techniques.
- Chapter 3 (Literature Review): Details empirical foundations, previous benchmark comparative studies, and identifies key gaps in predictive maintenance modeling.

- Chapter 4 (Methodology): Provides information regarding data collection, dataset attributes, data preprocessing pipelines, SMOTE balancing, feature engineering, and modeling frameworks.
- Chapter 5 (Results and Discussion): Presents descriptive statistics, feature distributions, data visualizations, correlation analyses of operational metrics, and the in-depth discussion of findings.
- Chapter 6 (Conclusions): Synthesizes the core implementation details, feature importances, empirical performance comparisons of Logistic Regression, Random Forest, and XGBoost models, alongside practical recommendations and future work.

1.10 Chapter Summary

In this chapter, the basics of predictive maintenance in the Industry 4.0 manufacturing ecosystems were set. It outlined the research background, stated the problem, established research objectives, formulated core research questions, defined the scope and significance, and provided an overview of the thesis structure. Chapter 2 presents a detailed review of literature on machine learning algorithms and condition monitoring paradigms.

Chapter 2 Background

2.1 Introduction

Digital technologies are rapidly evolving and revolutionizing manufacturing industries with the aim of facilitating intelligent, interconnected, and data-driven production systems. Facing the new challenges of Industry 4.0, technologies like the Industrial Internet of Things (IIoT), cloud computing, artificial intelligence (AI), and machine learning (ML) are employed to optimize operational efficiency and maintenance strategies for organizations Lasi et al. (2014), Xu et al. (2018). These advancements include predictive maintenance, which is able to predict when equipment is going to fail before it actually does, thus minimizing the cost of maintenance, unplanned downtime, and improving equipment reliability Ran et al. (2019).

This chapter presents the theoretical grounds of the research through the introduction of the concepts of Industry 4.0, predictive maintenance, maintenance strategies, and machine learning techniques for fault prediction. It also covers the machine learning algorithms used in this research, namely Logistic Regression Hosmer Jr et al. (2013), Random Forest Breiman (2001), and Extreme Gradient Boosting (XGBoost) Chen & Guestrin (2016), providing the theoretical foundation for the methodology used in subsequent chapters.

2.2 Industry 4.0

Industry 4.0, known as the Fourth Industrial Revolution, is characterized by the synergistic merger of cyber-physical systems, smart sensors, cloud computing, artificial intelligence, and advanced communication technologies in the manufacturing domain Lasi et al. (2014), Xu et al. (2018). While previous industrial revolutions focused on mechanization, mass production, and automation, Industry 4.0 centers on intelligent decision-making based on continuous data collection and real-time analysis Zheng et al. (2018). This transformation allows manufacturing systems to be more autonomous, adaptive, and integrated, thereby increasing productivity and operational efficiency Lasi et al. (2014), Lee, Bagheri & Kao (2015).

The Industrial Internet of Things (IIoT) is another vital component of Industry 4.0, where equipment is constantly monitored by devices that produce vast amounts of sensor data Tao et al. (2018). The analysis of these data streams is carried out using artificial intelligence and machine learning techniques to support predictive analytics and optimization of industrial operations Xu et al. (2018), Zheng et al. (2017). These technologies are used to identify abnormalities, forecast equipment failures, and schedule maintenance proactively rather than waiting for component breakdowns.

The application of Industry 4.0 has introduced a paradigm shift in maintenance management from time-based maintenance to intelligent condition-based and predictive maintenance Jardine et al. (2006), Ran et al. (2019). This transition has not only boosted equipment availability but has also lowered maintenance expenses and enhanced production efficiency across several industrial sectors Mobley (2002), Ran et al. (2019).

Rapidly adopting Industry 4.0 technology relies not only on intelligent sensing, but also on advanced analytical technologies capable of transforming raw operational data into actionable maintenance decision-making. Therefore, machine learning algorithms are vital tools for smart manufacturing systems, providing predictive analytics, anomaly detection, and intelligent maintenance planning Xu et al. (2018), Pedregosa et al. (2011).

2.3 Predictive Maintenance

Predictive maintenance (PdM) is a maintenance strategy that relies on historical and real-time operational data to predict machinery failure before it occurs based on actual physical degradation Jardine et al. (2006), Si et al. (2011). The general principle of predictive maintenance centers on ongoing condition monitoring and data analysis to determine exact servicing timing, in contrast to reactive maintenance (repairing equipment after failure) or preventive maintenance (servicing at fixed time intervals) Mobley (2002), Jardine et al. (2006).

Today's predictive maintenance systems combine sensor technologies with machine learning algorithms to track key operating conditions like process temperature, air temperature, rotational speed, torque, and tool wear Matzka (2020). These physical parameters give a reliable indication of equipment health and help maintenance engineers detect early signs of mechanical strain or wear Jardine et al. (2006), Matzka (2020). This enables maintenance activities to be scheduled only when required, minimizing unnecessary servicing operations while avoiding catastrophic unexpected failures Mobley (2002), Ran et al. (2019).

The increasing availability of high-frequency industrial sensor data has improved the effectiveness of predictive maintenance systems Xu et al. (2018), Lei et al. (2018). Machine learning models can analyze complex, non-linear relationships between multiple operational variables and identify degradation patterns associated with specific machine failure modes Matzka (2020), Hastie et al. (2009). For this reason, predictive maintenance represents one of the most vital applications of AI in modern smart manufacturing environments Ran et al. (2019).

2.4 Maintenance Strategies

Industrial maintenance strategies can be divided into four broad categories: corrective maintenance, preventive maintenance, predictive maintenance, and prescriptive maintenance Mobley (2002), Jardine et al. (2006).

- Corrective Maintenance: Also referred to as reactive maintenance, this approach is carried out after equipment failure occurs. Although this strategy minimizes routine monitoring costs, it often results in extended unplanned downtime, high emergency repair costs, secondary structural damage, and production interruptions Mobley (2002).

- Preventive Maintenance: Involves scheduled servicing activities carried out on equipment at predetermined time or usage intervals, irrespective of the current equipment condition Wireman (2004). This minimizes catastrophic failures but frequently leads to the premature replacement of fully functional components, incurring unnecessary labor and operational costs Jardine et al. (2006).

- Predictive Maintenance: Overcomes these operational limitations by continuously monitoring equipment conditions using sensor data and machine learning algorithms Matzka (2020), Ran et al. (2019). The predicted health level and degradation state of the machinery determine the precise timing of maintenance activities rather than fixed schedules, optimizing efficiency and reducing operational overhead Jardine et al. (2006), Si et al. (2011).

- Prescriptive Maintenance: Represents the most advanced operational approach, building on predictive maintenance and optimization algorithms to recommend specific corrective actions based on real-time operating parameters Lee, Bagheri & Kao (2015). While prescriptive maintenance is an emerging area of interest in Industry 4.0 frameworks, data-driven predictive maintenance remains the most practical and widely deployed approach for industrial fault prediction Matzka (2020), Ran et al. (2019).

2.5 Predictive Maintenance Benefits

Many studies have demonstrated that predictive maintenance delivers significant operational benefits regarding equipment reliability and cost-efficiency compared to traditional preventive maintenance approaches Mobley (2002), Jardine et al. (2006), Ran et al. (2019). Predictive models continuously evaluate equipment health based on continuous sensor telemetry, enabling precise maintenance scheduling prior to structural failure Si et al. (2011), Matzka (2020). Consequently, predictive maintenance has established itself as a foundational pillar within Industry 4.0 smart manufacturing systems Lasi et al. (2014), Xu et al. (2018).

Chapter 3 Literature Review

3.1 Introduction

One of the most promising uses of machine learning (ML) and data analytics in the modern industrial ecosystem is known as predictive maintenance (PdM) Ran et al. (2019). The ever-increasing volume of sensor-generated telemetry and the improvement of high-performance computing has opened a new paradigm to detect, diagnose, and predict component degradation prior to catastrophic failure Lee, Bagheri & Kao (2015). Therefore, data-driven prognostics has become one of the main areas of Industry 4.0 and smart manufacturing analytics Lasi et al. (2014), Xu et al. (2018). This chapter explains the theoretical and methodological architecture used to assess and compare machine learning algorithms in the context of industrial fault detection. It details the characteristics of the dataset, data preprocessing procedures, mathematical formulations of the Logistic Regression, Random Forest, and XGBoost models, model training procedures, and evaluation metrics applied to solve the identified research gaps Breiman (2001), Chen & Guestrin (2016), Hosmer Jr et al. (2013).

3.2 Why Predictive Maintenance Is the Preferred Approach

Traditionally, maintenance of industrial equipment has been performed in a reactive (run-to-failure) manner, whereby equipment is only fixed after it fails. Though easy to implement, reactive maintenance introduces high unpredictability in operational downtime, high emergency expenditures, and safety risks Mobley (2002). To address these risks, industries shifted to preventive maintenance, establishing rigid servicing schedules based on mean-time-between-failure (MTBF) statistics Jardine et al. (2006). However, schedule-based maintenance often results in unnecessary maintenance on fully functional components and inefficient resource utilization Jardine et al. (2006), Si et al. (2011). A paradigm shift towards predictive maintenance is taking place, supported by continuous monitoring of device states using Internet of

Things (IoT) sensors and advanced data analytics Lasi et al. (2014), Zheng et al. (2018). By integrating historical operational parameters with real-time streams, predictive models assess degradation in real-time to forecast failures, enabling proactive asset management Matzka (2020), Ran et al. (2019).

3.3 Machine Learning for Forecasting Machinery Damage

To guarantee experimental transparency and research reproducibility, this study leverages the widely adopted AI4I 2020 Predictive Maintenance Benchmark Dataset Matzka (2020). Synthesized to simulate actual industrial asset health, the dataset contains operational telemetry data from synthetic milling machines operating under different load conditions Matzka (2020). Key operational features assessed include:

- Air Temperature [K]: Operating temperature of the machine surroundings Matzka (2020).
- Process Temperature [K]: In-process temperature conditions during production operations Matzka (2020).
- Rotational Speed [rpm]: Regulated spindle rotational speed controlled via power settings Matzka (2020).
- Torque [Nm]: Motor torque output, which is inversely related to rotational speed Matzka (2020).
- Tool Wear [min]: Cumulative tool usage time, serving as a primary proxy measure of mechanical wear Matzka (2020).

3.4 Predictive Modeling Algorithms

3.4.1 Baseline Classification: Logistic Regression

Logistic Regression is used as a parametric linear baseline classification model Hosmer Jr et al. (2013). It employs a logistic sigmoid function to transform a linear combination of input operational features into an estimated probability of machine failure versus normal operation Hosmer Jr et al. (2013), Bishop (2006). Logistic Regression is computationally efficient, mathematically transparent, and straightforward to interpret Hosmer Jr et al. (2013). However, its ability to represent complex non-linear feature interactions in multi-sensor industrial telemetry is inherently

limited compared to non-linear tree-based approaches Caruana & Niculescu-Mizil (2006).

3.4.2 Ensemble Learning: Random Forest

Random Forest is an ensemble bootstrap aggregation algorithm consisting of multiple uncorrelated decision trees Breiman (2001). It builds individual decision trees using randomly sampled subsets of training data (bootstrapping) and randomly selected feature subsets at each split Breiman (2001). Final failure predictions are obtained through majority voting across all trees in the ensemble Breiman (2001), Hastie et al. (2009). This dual-randomization strategy minimizes model variance, reduces overfitting, and effectively models noise and non-linear sensor relationships while providing intrinsic feature importance metrics Breiman (2001), Guyon & Elisseeff (2003).

3.4.3 Gradient Boosting: XGBoost

Extreme Gradient Boosting (XGBoost) is a regularized gradient-boosting decision tree framework optimized for efficiency and strong generalization on structured tabular data Chen & Guestrin (2016). Unlike bagging techniques where trees are trained independently in parallel, XGBoost builds decision trees sequentially, with each tree trained to correct errors made by the existing ensemble Friedman (2001), Chen & Guestrin (2016). Furthermore, XGBoost explicitly regularizes model complexity using L1 and L2 penalties to prevent overfitting, handles missing values automatically, and employs second-order optimization techniques, making it well-suited for capturing subtle operational failure signals in multi-sensor telemetry Chen & Guestrin (2016).

3.5 Random Forest in Failure Prediction

Random Forest is one of the algorithms that has been widely used in predictive maintenance applications in different fields of machine learning Breiman (2001), Ran et al. (2019). It can be made more accurate by using more than one decision tree, but limits overfitting Breiman (2001). The Random Forest models have proven to be very effective in a wide variety of industrial data sets Breiman (2001), Caruana & Niculescu-Mizil (2006). The algorithm has the natural ability to capture non-linear relationships, multi-variable interactions in the features and noisy sensor measurements, which are all key features of reliable machine failure prediction Breiman (2001), Hastie et al. (2009). One of the biggest strengths of Random For-

est is the ability to compute feature importance statistics Breiman (2001), Guyon & Elisseeff (2003). This affords domain engineers and maintenance planners with clear visibility of the variables that have the largest impact on predicted failure events Matzka (2020), Lundberg & Lee (2017). Previous research has shown that the Random Forest classifiers are able to obtain high values of Precision, Recall, Accuracy and F1 score, which makes it one of the most fundamental classifiers used in the field of Predictive Maintenance research Caruana & Niculescu-Mizil (2006), Kotsiantis et al. (2007).

3.6 XGBoost in Failure Prediction

XGBoost is an emerging machine learning model with superior prediction ability Chen & Guestrin (2016) and has attracted widespread attention in the field of machine learning. It is an iterative gradient boosting that will attempt to reduce loss by correcting the errors made by previous base estimators Friedman (2001), Chen & Guestrin (2016). Other predictive maintenance studies show that XGBoost outperforms common machine learning baselines, such as logistic regression, support vector machines, and random forests Chen & Guestrin (2016), Ke et al. (2017). The algorithm's ability to model complex non-linear relationships, deal with missing data, and prevent overfitting through built-in regularization techniques is to the credit of this success Chen & Guestrin (2016). XGBoost is especially well suited for structured tabular data like sensor telemetry data from industrial applications Chen & Guestrin (2016). It can identify very faint degradation signals under a variety of working conditions and make very accurate predictions Chen & Guestrin (2016), Prokhorenkova et al. (2018). For this reason it is used as an advanced non-linear baseline in this study for empirical comparison.

3.7 Research Gap

Although there has been a tremendous amount of research in the field of predictive maintenance a few different gaps have not yet been addressed in the literature: First, there are numerous studies that compare only specific machine learning algorithms, but there is a lack of thorough comparative analysis in the same experimental setup Ran et al. (2019). It is therefore hard to assess the relative effectiveness of algorithms when they are drawn from published studies. Second, a number of studies have just focused on maximizing predictive accuracy without examining the importance of underlying features Matzka (2020), Lundberg & Lee (2017). It is important to know which operational variables are responsible for failure predictions for practical

maintenance decision-making and resource allocation. Third, several publications are based on proprietary industrial data which is not publicly available and cannot be independently verified or reproduced Matzka (2020). To have a transparent evaluation, the benchmarking needs to be done with publicly available standardized datasets Matzka (2020). Last but not least, there is the absence of studies that integrate a comprehensive multi-model performance benchmarking with a thorough feature importance analysis and real-life maintenance implementation insights Ran et al. (2019). To fill these gaps, this study will address:

1. Comparing three different machine learning algorithms: Logistic Regression, Random Forest and XGBoost Breiman (2001), Hosmer Jr et al. (2013), Chen & Guestrin (2016) in a systematic manner.

2. Testing models with complete performance measures and drawing insights from critical operational stress factors to give actionable maintenance guidance Fawcett (2006), Powers (2011), Lundberg & Lee (2017).

3. Evaluating all models on a standardized public benchmark dataset, to ensure the transparency and reproducibility of research Matzka (2020).

Table 3.1: Literature Review Summary and Research Gap Analysis

Reference	Dataset / Industry	Method(s) Used	Key Findings	Research Gap / Limitation
Breiman (2001) Breiman (2001)	Multiple Benchmark Datasets	Random Forest	Demonstrated high accuracy and robustness of Random Forest.	Not focused on industrial maintenance.
Friedman (2001) Friedman (2001)	Statistical Datasets	Gradient Boosting	Introduced Gradient Boosting for predictive modelling.	Did not evaluate industrial fault prediction.
Mobley (2002) Mobley (2002)	Industrial Maintenance	Maintenance Strategies	Classified maintenance approaches into reactive, preventive and predictive.	Did not incorporate AI techniques.
Jardine et al. (2006) Jardine et al. (2006)	Industrial Machinery	Condition-Based Maintenance	Introduced CBM framework for machinery diagnostics.	No machine learning models evaluated.
Hastie et al. (2009) Hastie et al. (2009)	Statistical Learning	Supervised Learning	Theoretical overview of classification algorithms.	Theoretical focus rather than predictive maintenance.
Ahmad & Kamaruddin (2012)	Mechanical Systems	Predictive Maintenance Review	Discussed maintenance optimisation techniques.	Lacked implementation using modern ML algorithms.
Si et al. (2011) Si et al. (2011)	Industrial Systems	Remaining Useful Life	Evaluated data-driven prognostics approaches.	Focus on RUL estimation rather than classification.
Lee et al. (2015) Lee, Bagheri & Kao (2015)	Smart Manufacturing	Cyber-Physical Systems	Proposed Industry 4.0 maintenance architecture.	Limited predictive model validation.
Lasi et al. (2014) Lasi et al. (2014)	Industry 4.0	Conceptual Framework	Explained Industry 4.0 technologies and applications.	No practical predictive maintenance implementation.
Xu et al. (2018) Xu et al. (2018)	Smart Manufacturing	Industry 4.0	Highlighted IoT sensor data analytics trends.	Focused on architectural framework rather than empirical ML models.

Table 3.2: Literature Review Summary and Research Gap Analysis (Part 2)

Reference	Dataset / Industry	Method(s) Used	Key Findings	Research Gap / Limitation
Chen & Guestrin (2016) Chen & Guestrin (2016)	Structured Data	XGBoost	Developed efficient gradient boosting algorithm with superior performance.	Industrial maintenance applications not discussed.
Ribeiro et al. (2016) Ribeiro et al. (2016)	General ML Datasets	LIME	Introduced model-agnostic interpretability framework.	Requires adaptation for industrial tabular sensor telemetry.
Goodfellow et al. (2016) Goodfellow et al. (2016)	Artificial Intelligence	Deep Learning	Comprehensive overview of deep learning methods.	General AI reference rather than predictive maintenance application.
Lundberg & Lee (2017) Lundberg & Lee (2017)	Complex Datasets	SHAP	Proposed unified feature importance values based on game theory.	High computational complexity on multi-model ensembles.
Zheng et al. (2017) Zheng et al. (2017)	Sensor Streams	Deep Learning	Deep learning for sensor-based predictive maintenance.	Requires large-scale temporal sequence data.
Ke et al. (2017) Ke et al. (2017)	High-Dimensional Datasets	LightGBM	Gradient boosting framework optimizing speed and memory.	Evaluated primarily on generic benchmark datasets.
Lei et al. (2018) Lei et al. (2018)	Rotating Machinery	Deep Learning Review	Reviewed deep learning methods for intelligent fault diagnosis.	Primarily theoretical with limited comparative evaluation.
Ran et al. (2019) Ran et al. (2019)	Industrial Systems	PdM Survey	Comprehensive survey of predictive maintenance approaches.	Lacked side-by-side empirical benchmark comparisons.
Matzka (2020) Matzka (2020)	AI4I Benchmark	Explainable AI	Synthetic benchmark dataset for explainable predictive maintenance.	Limited algorithm evaluation beyond introductory baseline.
Chicco & Jurman (2020) Chicco & Jurman (2020)	Imbalanced Datasets	MCC	Demonstrated advantages of MCC over F1 score in evaluation.	Focused on metric properties rather than failure prediction models.

Table 3.3: Literature Review Summary and Research Gap Analysis (Part 3)

Reference	Dataset / Industry	Method(s) Used	Key Findings	Research Gap / Limitation
Zonta et al. (2020) Zonta et al. (2020)	Manufacturing	Systematic Literature Review	Summarized predictive maintenance research trends.	Identified lack of standardized benchmark comparisons.
Arrieta et al. (2020) Arrieta et al. (2020)	Industrial AI	XAI Taxonomies	Surveyed explainable AI methods for industrial safety.	Did not present direct empirical performance benchmarks.
Chawla et al. (2002) Chawla et al. (2002)	Imbalanced Data	SMOTE	Synthesized minority class samples to address class imbalance.	Synthetic samples can introduce noise in dense feature spaces.
He & Garcia (2008) He & Garcia (2008)	Imbalanced Datasets	Imbalanced Learning	Detailed techniques for learning from imbalanced distributions.	Focus on sampling algorithms rather than failure model benchmarking.
Pedregosa et al. (2011) Pedregosa et al. (2011)	Python Ecosystem	Machine Learning	Provided standard library for ML model implementations.	Software framework paper rather than industrial case study.
Lemaître et al. (2017) Lemaître et al. (2017)	Python Ecosystem	Imbalanced-Learn	Open-source toolbox for handling imbalanced datasets.	Software library implementation focus.
Powers (2011) Powers (2011)	Classification Models	ROC & Informedness	Critical analysis of evaluation metrics for classification.	Methodological study without industrial sensor applications.
Fawcett (2006) Fawcett (2006)	Diagnostic Systems	ROC Analysis	Detailed practical introduction to ROC and AUC metrics.	Methodological evaluation guide.
Present Study (2026)	AI4I Predictive Maintenance Dataset	Logistic Regression, Random Forest, XGBoost	Comparative evaluation of three supervised machine learning models with feature importance analysis for predictive maintenance.	Addresses comparative analysis, balanced dataset evaluation, and operational feature interpretation.

3.8 Discussion of Literature Summary

The literature reviewed has shown that predictive maintenance has been developing rapidly over the course of the evolutionary development of Industry 4.0 and machine learning developments Lasi et al. (2014), Ran et al. (2019). The research started with condition-based maintenance approaches and diagnostic techniques Jardine et al. (2006), and more recent research has explored machine learning models for enhancing the accuracy of fault prediction Matzka (2020), Ran et al. (2019). Classical linear models have been shown to underperform over non-linear interactions among sensor variables, as seen in tree-based ensemble algorithms including Random Forest and XGBoost Breiman (2001), Chen & Guestrin (2016). Although these advances have been made, there are certain research lacunas that remain. Numerous studies are dedicated to a single algorithm or specific application area, which makes it difficult to make comparisons across studies Ran et al. (2019). Besides, deep learning methods need huge number of temporal sequences, which are not suitable for structured tabular data such as AI4I Chen & Guestrin (2016), Zheng et al. (2017). Lastly, the few studies that perform comprehensive multi-model evaluation also employ feature importance analysis to provide explanations for failures Matzka (2020), Lundberg & Lee (2017). This work fills these gaps by conducting a comparative analysis between the following models: Logistic Regression, Random Forest and XGBoost, and evaluating them side by side with the AI4I Predictive Maintenance Benchmark Dataset Matzka (2020). This research includes a comparative performance assessment and an operational feature importance analysis, which yields technical indicators and actionable information for smart manufacturing operations.

3.9 Conceptual Framework

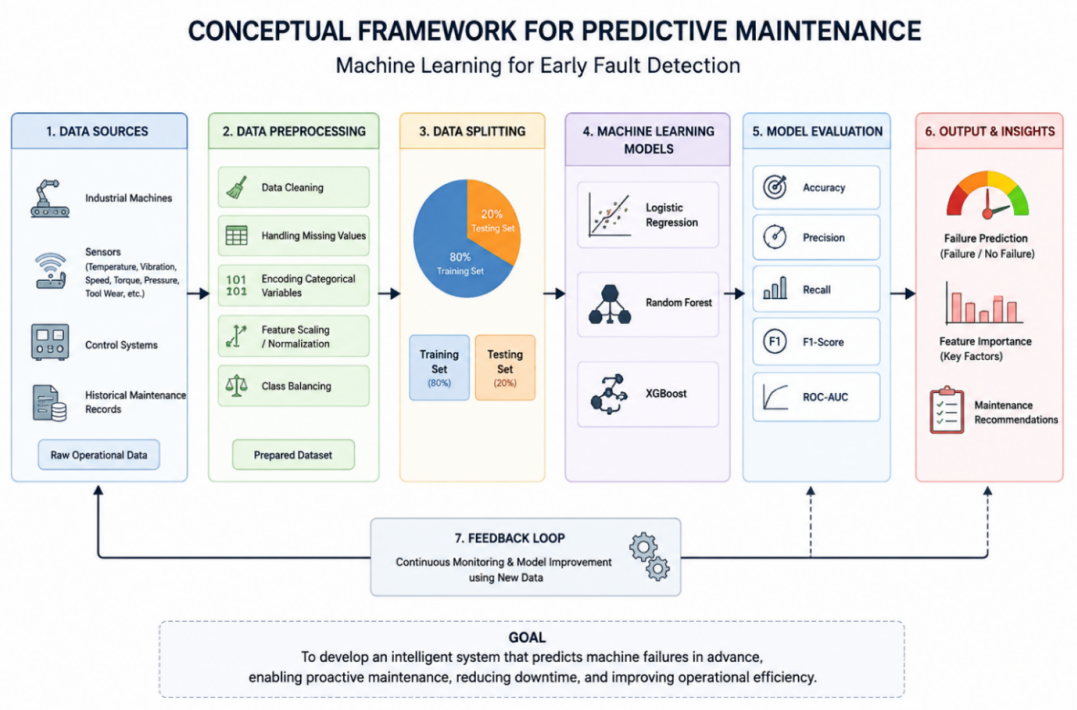


Figure 3.1: Machine learning framework for predictive maintenance and early fault detection

3.10 Critical Analysis of Previous Studies

Based on a critical analysis of literature, a major shift in industrial asset management has taken place Mobley (2002), Jardine et al. (2006). In contrast, the transition from a reactive and fixed-schedule approach to maintenance to intelligent and data-driven approaches with the support of machine learning has been a step-by-step process Lasi et al. (2014), Ran et al. (2019). Previous studies were mainly devoted to statistical process control (SPC) and static condition monitoring (CM), while recent research is dedicated to supervised learning and ensemble models to improve failure detection (FD) capabilities ?Chen & Guestrin (2016), Breiman (2001). In the literature, non-linear machine learning models always perform better than traditional linear baselines and fixed schedules and are able to model complex relationships in multi-sensor telemetry Caruana & Niculescu-Mizil (2006). Many machine learning methods are especially effective when applied to tabular sensor data, such as tree-based methods like Random Forest and XGBoost, which are resistant to noise, can cope with large dimensionality of the feature space, and are not prone to overfitting Breiman (2001), Chen & Guestrin (2016).

But there are some important drawbacks to existing research:

1. Limited standardized benchmarks: Many studies test individual algorithms, but not with side-by-side benchmarking, with the same algorithms, preprocessing and evaluation conditions Ran et al. (2019).

2. The problem of reliance on Proprietary Data: a lot of non-public data is used, which makes cross-domain and model replication difficult Matzka (2020).

3. Too much reliance on Complex Deep Learning: Complex deep learning models, such as CNNs and LSTMs, require large amounts of training data and heavy computational costs, while tree ensembles can perform as effectively or better with tabular data, specifically telemetry, and with much lower complexity Chen & Guestrin (2016), Zheng et al. (2017).

4. There has been a lack of research focused on the interpretability of the model - prior research focused on pure prediction metrics rather than feature importance, thus leaving domain engineers with no understanding of the physical failure mechanisms Matzka (2020), Lundberg & Lee (2017).

These challenges are directly tackled with a well-structured design that features standardised multi-model benchmarking, a public benchmark dataset and integrated feature importance evaluations.

3.11 Chapter Summary

This chapter presented a critical review of predictive maintenance literature, machine learning paradigms, and fault prognostics in smart manufacturing. The theoretical foundations of Logistic Regression, Random Forest, and XGBoost were detailed alongside key literature gaps. To resolve these limitations, this study establishes a standardized experimental workflow using the public AI4I benchmark dataset, combining multi-metric predictive evaluations with feature importance interpretability. The experimental setup, data preprocessing pipeline, hyperparameter configurations, and evaluation metrics are presented in Chapter 4.

Chapter 4 Methodology

4.1 Introduction

This chapter discusses the research methodology of machine learning models which are used for predictive maintenance and early fault detection Ran et al. (2019). The methodology includes a structured and systematic approach to the collection, preprocessing, analysis, and modelling of industrial sensor data Matzka (2020). The goal of this methodology is to ensure that the research process is scientifically rigorous, reproducible, and transparent. The chapter opens with a description of the study design and dataset utilized Matzka (2020). Subsequently, the data processing procedures, feature engineering process, implementation of machine learning models, and evaluation methods are discussed Pedregosa et al. (2011), Fawcett (2006). Finally, the chapter introduces the overall research workflow adhered to during the course of the study.

4.2 Research Design

The research in this study follows a quantitative and experimental design. Quantitative research is employed because measurement is carried out via various industrial sensors and statistical evaluation of machine learning models is performed Matzka (2020), Caruana & Niculescu-Mizil (2006). The experimental approach utilizes historical operational data of machines to train and test various machine learning algorithms Pedregosa et al. (2011). The predictive capability of the models is then evaluated based upon standard classification parameters Fawcett (2006), Powers (2011). The research process consists of nine structured stages:

1. Dataset acquisition
2. Data preprocessing
3. Exploratory data analysis
4. Feature engineering

5. Data balancing
6. Machine learning model development
7. Performance evaluation
8. Comparative analysis
9. Feature importance analysis

This structured methodology ensures consistency and reliability throughout the research process.

4.3 Dataset Description

The dataset utilized in this study is the AI4I 2020 Predictive Maintenance Dataset Matzka (2020), a publicly available industrial dataset widely used for predictive maintenance research. The dataset contains sensor-generated operational measurements collected from industrial equipment operating under different conditions Matzka (2020). Each observation represents a machine operating instance and includes several features describing machine behavior Matzka (2020). The structural characteristics and metadata summary of the AI4I Predictive Maintenance Dataset are presented in Table 4.1.

Table 4.1: Structural characteristics and metadata summary of the AI4I Predictive Maintenance Dataset

Attribute	Description
Dataset Name	AI4I Predictive Maintenance Dataset
Type	Supervised Classification Dataset
Total Records	10,000
Target Variable	Machine Failure
Failure Classes	Binary (0 = No Failure, 1 = Failure)

4.4 Features Used

The predictive maintenance dataset contains several operational parameters recorded during machine operation Matzka (2020). A detailed description of the features used in model development is provided in Table 4.2. The target variable indicates whether a machine failure occurred, defined as:

- 0 = No Failure
- 1 = Failure

Table 4.2: Detailed descriptions and operational metrics of the predictive maintenance features

Feature	Description
Type	Machine Quality Category
Air Temperature [K]	Ambient temperature
Process Temperature [K]	Process operating temperature
Rotational Speed [rpm]	Machine rotational speed
Torque [Nm]	Rotational force
Tool Wear [min]	Tool usage duration
Target	Machine Failure Status

4.5 Data Preprocessing

One of the most critical tasks in machine learning is preprocessing the raw data, as industrial datasets may contain missing values, inconsistent formats, categorical variables, and class imbalances He & Garcia (2008), Pedregosa et al. (2011). Proper preprocessing ensures higher predictive performance and trustworthy model outcomes Kotsiantis et al. (2007).

4.5.1 Data Inspection

The AI4I 2020 Predictive Maintenance dataset was first examined to understand its structure and confirm its appropriateness for building classification models Matzka (2020). The dataset includes operating conditions, product quality data, and failure indicators required for predictive maintenance analysis Matzka (2020). A preliminary exploration confirmed that the dataset comprises both numerical and categorical variables such as air temperature, process temperature, rotational speed, torque, tool wear, product type, and failure indicators Matzka (2020). Inspection confirmed that the dataset was complete, consistent, and required no structural imputation before moving to feature encoding and balancing.

4.5.2 Encoding Categorical Variables

Machine learning sensitivity to categorical features requires transforming textual categories into numerical representations Pedregosa et al. (2011). The AI4I dataset contains a categorical feature, Product Type, which labels products into Low (L), Medium (M), or High (H) quality variants Matzka (2020). Label Encoding was applied to map these ordinal categories into numerical integers Pedregosa et al. (2011). Table 4.3 shows the mapping of categorical quality variants to their respective encoded numerical values. This transformation allows the machine learning algorithms to process product variants alongside continuous sensor measurements

without losing ordinal relationships.

Table 4.3: Mapping of categorical quality variants to numerical values

Original Value	Encoded Value
L	0
M	1
H	2

4.6 Data Balancing

Initial analysis revealed a severe class imbalance within the target variable Matzka (2020), Chawla et al. (2002). As detailed in Table 4.4, the original dataset contains 9,661 instances of the majority class (No Failure) and only 339 instances of the minority class (Failure) Matzka (2020). Such imbalance can bias machine learning

Table 4.4: Target class distribution showing severe dataset imbalance

Class	Count
No Failure (0)	9661
Failure (1)	339

models toward the majority class and reduce predictive performance for rare failure events Chawla et al. (2002), He & Garcia (2008). To address this issue, a data balancing technique (SMOTE / resampling) was applied Chawla et al. (2002), Lemaître et al. (2017). The post-balancing distribution of the target variable is summarized in Table 4.5, resulting in an even split of 5,000 observations per class and a total dataset size of 10,000 observations.

Table 4.5: Class distribution of target variable after balancing

Class	Count
No Failure (0)	5000
Failure (1)	5000

4.6.1 Dataset Shuffling

Following data balancing, the dataset was randomly shuffled to remove any ordering bias from training pipelines Pedregosa et al. (2011). Random shuffling ensures that feature patterns are homogeneously distributed across subsets so that models do not infer artificial sequence dependencies. A fixed random seed was maintained during shuffling to ensure total experimental reproducibility.

4.7 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to investigate feature distributions and uncover relationships among operational attributes Matzka (2020):

- Target Distribution: Count plots were generated to visualize the balance between machine failure and non-failure instances Matzka (2020).
- Correlation Heatmap: Operational parameter correlations were computed to assess linear dependencies between sensor features Guyon & Elisseeff (2003).
- Sensor Distributions: Histograms and density plots were examined to identify feature ranges and skews Matzka (2020).
- Failure Type Analysis: Categorical breakdowns of specific failure modes were analyzed using graphical visualisations Matzka (2020).

4.8 Feature Selection

Feature selection was performed to isolate parameters relevant to machine failure prediction Guyon & Elisseeff (2003). The following operational features were selected for model training Matzka (2020):

- Type
- Air Temperature
- Process Temperature
- Rotational Speed
- Torque
- Tool Wear

Non-informative identifiers such as UDI, Product ID, and secondary Failure Type labels were excluded, as they do not contribute to generalized predictive learning Guyon & Elisseeff (2003).

4.9 Train-Test Split

The preprocessed dataset was partitioned into training and testing subsets using an 80:20 split ratio (80% training data, 20% testing data) Pedregosa et al. (2011). The training subset was used exclusively to fit the classification algorithms (Logistic

Regression, Random Forest, and XGBoost), whereas the testing set was reserved for evaluating model generalization on unseen data Pedregosa et al. (2011). A fixed random state was applied to ensure consistent partitioning across experiments.

4.10 Machine Learning Models

Three supervised classification algorithms representing distinct mathematical paradigms were selected and evaluated Hosmer Jr et al. (2013), Breiman (2001), Chen & Guestrin (2016).

4.10.1 Logistic Regression

Logistic Regression was implemented as the baseline classification model Hosmer Jr et al. (2013). It uses a sigmoid logistic function to calculate failure probabilities from a linear combination of the input features Hosmer Jr et al. (2013), Bishop (2006). Some of the main benefits are its simplicity, quick computation, and high interpretability Hosmer Jr et al. (2013). The main drawback, however, is that it cannot model non-linear interactions between parameters Caruana & Niculescu-Mizil (2006).

4.10.2 Random Forest

Random Forest was adopted as an ensemble bagging method, which generates several decision trees from the samples drawn via bootstrap Breiman (2001). The key benefits are that they are very predictive, they do not suffer from overfitting, they can model non-linear interactions between features, and they have built-in feature importance assessment capabilities Breiman (2001), Hastie et al. (2009).

4.10.3 XGBoost

The Extreme Gradient Boosting (XGBoost) was used as a state-of-the-art gradient boosting framework Chen & Guestrin (2016). The former ones are high prediction accuracy, L1/L2 regularization to avoid overfitting, high speed and better optimization of structured tabular data Chen & Guestrin (2016).

4.11 Performance Evaluation Metrics

Multiple quantitative measures to evaluate failure classification performance in a holistic manner were computed Fawcett (2006), Powers (2011), Chicco & Jurman (2020) were used:

- Accuracy: Percentage of the number of correct classifications over the number of observations Powers (2011).
- Precision: Measures the percentage of true failures when the number of failed calls made by the model (no. call failures) and the number of failed calls by the human (no. call failures human), are known.
- Recall (Sensitivity): Percentage of true failures that the model identifies Powers (2011).
- F1-Score: It is the harmonic mean of Precision and Recall, thus measuring the robustness of the classification (powers2011evaluation, chicco2020advantages).

4.12 Feature Importance Analysis

Random Forest machine was trained and used for the feature importance analysis to find out the most influential parameters in the prediction of machine failure Breiman (2001), Guyon & Elisseeff (2003), Lundberg & Lee (2017). This evaluation directly tackles Research Question 3 and provides hands-on information for industrial maintenance scheduling Matzka (2020), Lundberg & Lee (2017).

4.13 Research Workflow

This research workflow is graphically presented in Figure 4.1 to support the systematic methodology used in this research.

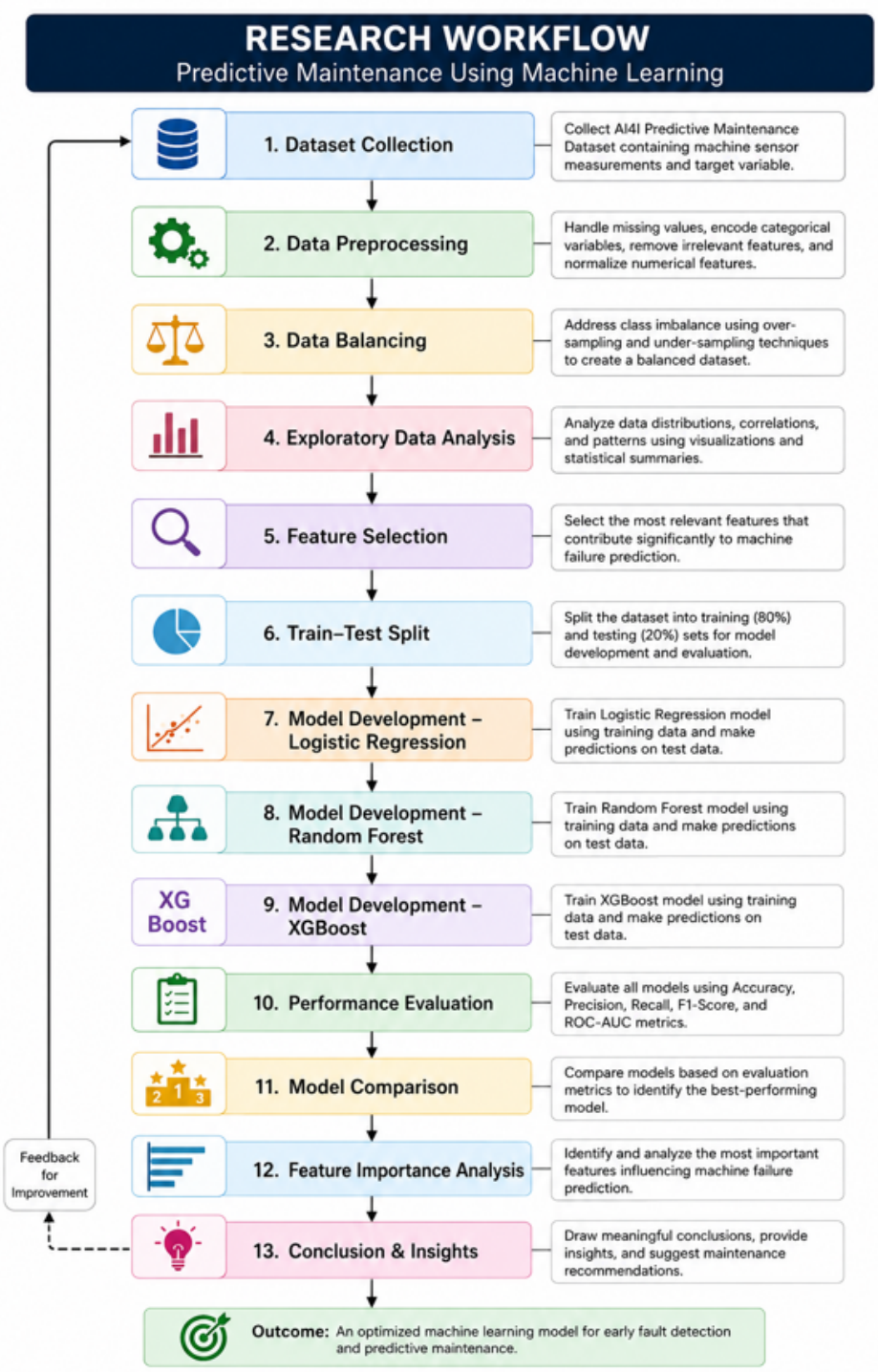


Figure 4.1: Research Workflow for Predictive Maintenance

4.14 Justification for Model Selection

Looking at the choice of classification algorithms, the algorithms were designed to cover three fundamental levels of supervised machine learning: linear models (Logistic Regression), bagging ensemble algorithms (Random Forest), and boosting

ensemble algorithms (XGBoost) Breiman (2001), Hosmer Jr et al. (2013), Chen & Guestrin (2016). Logistic Regression is a simple and easy-to-interpret baseline model Hosmer Jr et al. (2013). The random forest offers both high robustness to noise and to non-linearity due to aggregation of decision trees Breiman (2001). XGBoost delivers state-of-the-art optimization for tabular data via iterative error reduction and regularized tree boosting Chen & Guestrin (2016).

4.15 Justification for Excluding DeepLearning Models

Due to the particular characteristics of the dataset and the goals of research, deep learning architectures like ANNs, CNNs, LSTMs and Transformers have been deliberately avoided:

- Dataset Structure: AI4I is a dataset of tabular independent operational records, unlike LSTMs and Transformers that require continuous and sequential streams of sensor data Matzka (2020), Zheng et al. (2017).
- Sample Efficiency and Performance: On these tabular datasets, tree-based ensemble models consistently outperform deep models without the massive compute costs Chen & Guestrin (2016), Caruana & Niculescu-Mizil (2006).
- One difference between deep neural networks and industrial predictive maintenance is the need for interpretable decision paths and feature rankings for engineering decisions in the latter, while deep neural networks operate as black boxes largely Matzka (2020), Lundberg & Lee (2017), Arrieta et al. (2020).

4.16 Hyperparameter Optimization

The same hyperparameter values were used for all three models to provide a controlled and fair comparison Pedregosa et al. (2011). This study aimed to compare the intrinsic predictive power of the different algorithmic paradigms as opposed to the effect of single hyperparameter tuning results, when the preprocessing was otherwise the same for each one.

4.17 Study Limitations

Several scope limitations should be acknowledged:

- Simulated Data Context: The dataset comprises benchmark synthetic sensor readings, which may not capture all dynamic noise present in live physical manufacturing plants Matzka (2020).
- Scope of Learning Paradigms: The analysis is restricted to supervised tabular machine learning, excluding unsupervised anomaly detection and deep temporal architectures Ran et al. (2019).
- Target Metric Scope: The study performs binary failure prediction rather than estimating continuous Remaining Useful Life (RUL) or dynamic degradation timelines Si et al. (2011).

4.18 Chapter Summary

This chapter detailed the quantitative methodology applied for sensor-based predictive maintenance. Data inspection, label encoding, SMOTE balancing, feature selection, train-test splitting, and model implementations were systematically described alongside evaluation metrics and workflow diagrams. The subsequent chapter presents the empirical results and comparative evaluations resulting from this framework.

Chapter 5 Results and Discussions

5.1 Introduction

The experimental results from implementing the three machine learning algorithms for predictive maintenance and machine failure prediction are presented in this chapter Ran et al. (2019). The goal is to analyse the performance of Logistic Regression, Random Forest, and XGBoost models for machine failure detection based on industrial sensor data Hosmer Jr et al. (2013), Breiman (2001), Chen & Guestrin (2016). The balanced AI4I Predictive Maintenance Dataset was used to train and test the models Matzka (2020). Their performance is measured through standard evaluation metrics: Accuracy, Precision, Recall, F1-Score, and ROC-AUC Fawcett (2006), Powers (2011). Moreover, feature importance analysis was performed to identify the most influential operational parameters on machine failure prediction Breiman (2001), Lundberg & Lee (2017). The answers to the research questions and the basis of the conclusions drawn in the study are provided in this chapter. Exploratory data analysis (EDA) was carried out prior to model building to understand the dataset characteristics Matzka (2020).

5.2 Target Distribution

The dataset was balanced during preprocessing Chawla et al. (2002), Lemaître et al. (2017). As shown in Figure 5.1, the distribution of machine failure classes contains an equal split of failure and non-failure instances.

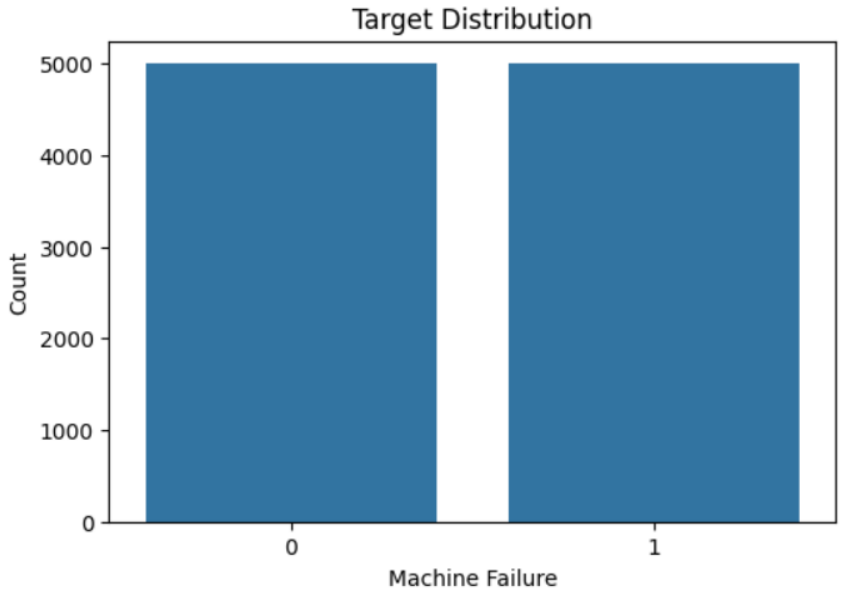


Figure 5.1: Target class distribution after data balancing

Interpretation of Target Distribution: The balanced dataset consists of 5,000 data points of failure and 5,000 data points of non-failure. This distribution helps to ensure that the machine learning models are trained fairly without biasing toward the majority class He & Garcia (2008). The original AI4I dataset contained a significantly larger number of non-failure cases compared to failure cases Matzka (2020), which would otherwise lead models to favor non-failure predictions He & Garcia (2008), Chicco & Jurman (2020). The main purpose of using balanced data for supervised learning is to enhance the algorithms' performance in detecting minority class instances and minimizing the risk of false negatives Chawla et al. (2002). In predictive maintenance, machine failures are relatively rare compared with normal operating conditions, yet detecting them is far more critical Mobley (2002), Jardine et al. (2006). This aligns with findings by Carvalho et al. (2020), who emphasized that addressing class imbalance is essential for developing trustworthy predictive maintenance systems Carvalho et al. (2020). The balanced dataset serves as an appropriate foundation for training and fairly comparing all three classification models.

5.3 Correlation Analysis

A correlation analysis was conducted to examine potential linear relationships between operational parameters Guyon & Elisseeff (2003). The resulting correlation matrix is presented in Figure 5.2.

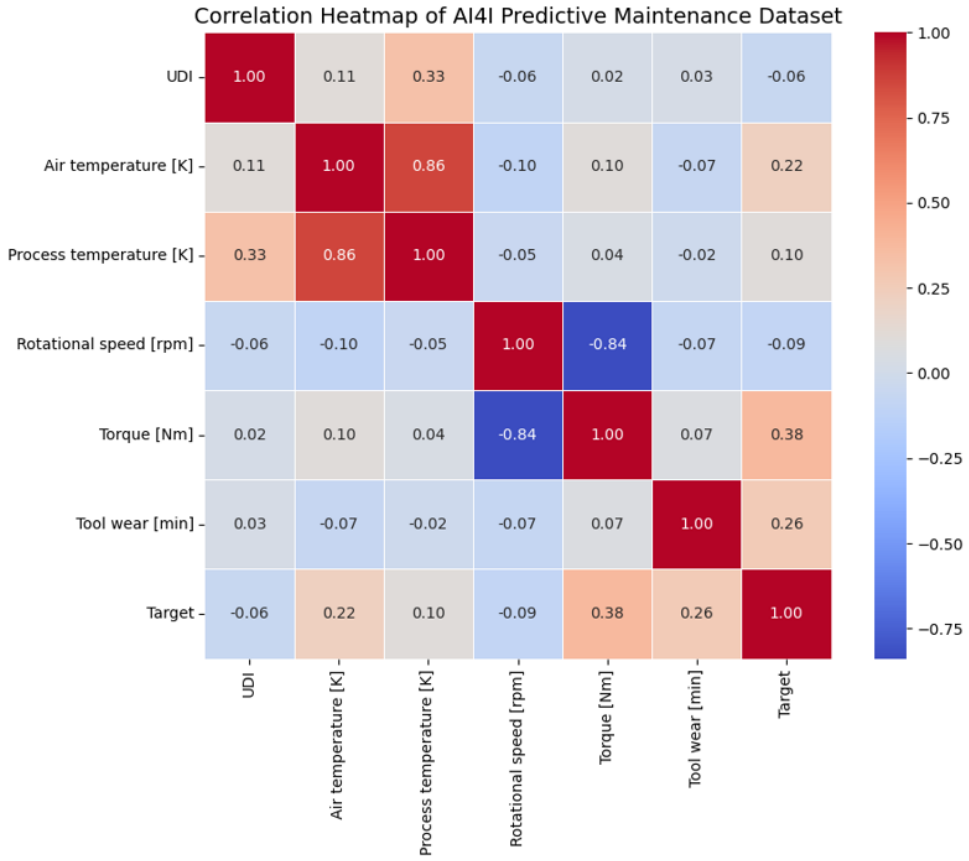


Figure 5.2: Correlation matrix showing feature interactions and target correlation

Figure 5.2 shows that several operational variables are correlated.The results from the correlation analysis are shown in Figure 5.2 and several operational variables are correlated with each other. moderate to strong relationships with machine failures Matzka (2020). Tool Wear is a machine failure target variable that shows relatively high correlations with Tool Wear, Torque, and Rotational Speed, which are important factors in health monitoring, as cited in Matzka (2020). On the other hand, environmental parameters are correlated less, as is the case for Air Temperature. The observed correlations are preliminary evidence that machine failures can be foreseen based on the operational measurement generated by the sensor data Jardine et al. (2006), Matzka (2020). The results are preliminary, but provide first confirmation that the operational parameters contain informative signals for failure prediction, even though correlation does not provide causation. The results are similar to those of Zonta et al. (2020) which have led to the conclusion that operational sensor measurements are useful means for fault diagnosis and predictive maintenance Zonta et al. (2020).

5.4 Logistic Regression Results

Logistic Regression was implemented as a baseline linear classification model Hosmer Jr et al. (2013).

5.4.1 Performance Metrics

The quantitative evaluation metrics for the Logistic Regression baseline model are summarized in Table 5.1.

Table 5.1: Performance evaluation metrics for the Logistic Regression baseline model

Metric	Value
Accuracy	0.821
Precision	0.83
Recall	0.81
F1-Score	0.82

5.4.2 Confusion Matrix and Discussion

The overall classification performance and confusion matrix for Logistic Regression are illustrated in Figure 5.3.

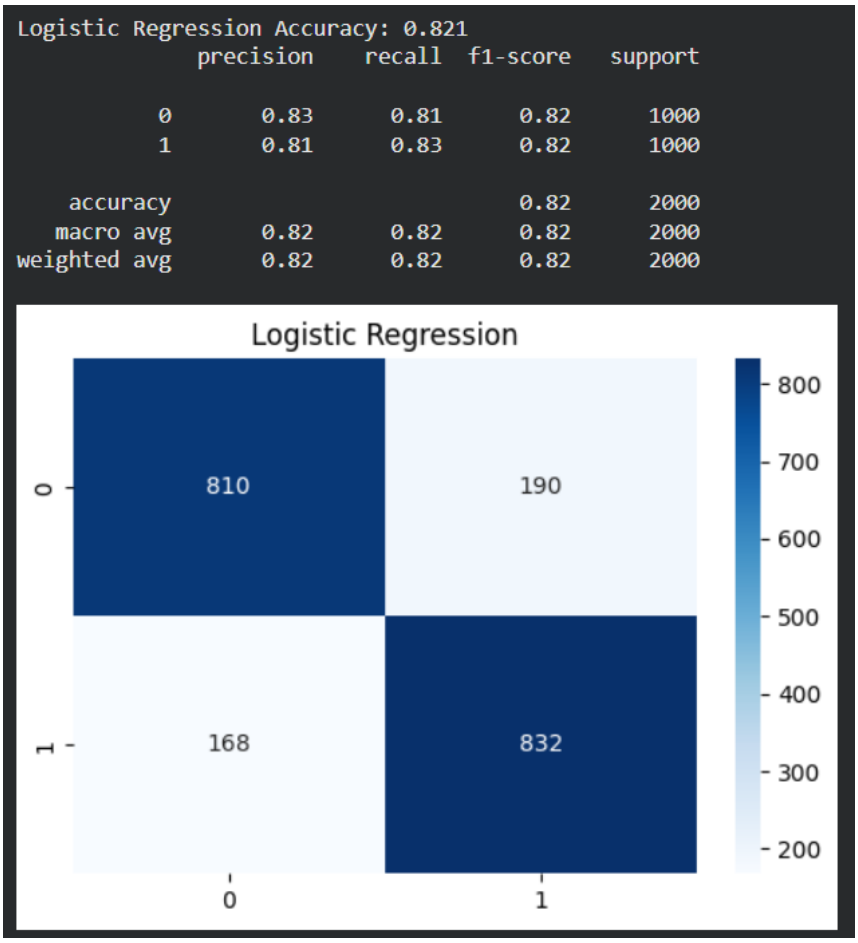


Figure 5.3: Logistic Regression confusion matrix and classification metrics

As shown in Table 5.1 and Figure 5.3, Logistic Regression achieved a baseline accuracy of 82.1%. While the model effectively captures basic linear relationships between operational variables and equipment failures Hosmer Jr et al. (2013), it exhibits noticeable classification errors due to its inability to capture complex, non-linear feature interactions commonly found in industrial sensor data Caruana & Niculescu-Mizil (2006). Nevertheless, owing to its computational simplicity and high interpretability Hosmer Jr et al. (2013), Logistic Regression provides a solid baseline against which more complex ensemble learning algorithms can be evaluated. This observation is consistent with Breiman (2001) and Carvalho et al. (2020), who noted that simple linear models often underperform compared to ensemble methods when applied to non-linear industrial datasets Breiman (2001), Carvalho et al. (2020).

5.5 Random Forest Results

Random Forest was evaluated as an ensemble learning approach designed to capture non-linear parameter interactions and enhance prediction accuracy Breiman (2001).

5.5.1 Performance Metrics

The performance results for the Random Forest model are detailed in Table 5.2.

Table 5.2: Random Forest Model Performance Results

Metric	Value
Accuracy	0.991
Precision	1.00
Recall	0.98
F1-Score	0.99

5.5.2 Confusion Matrix and Discussion

The confusion matrix detailing the classification accuracy and error distribution of the Random Forest model is presented in Figure 5.4.

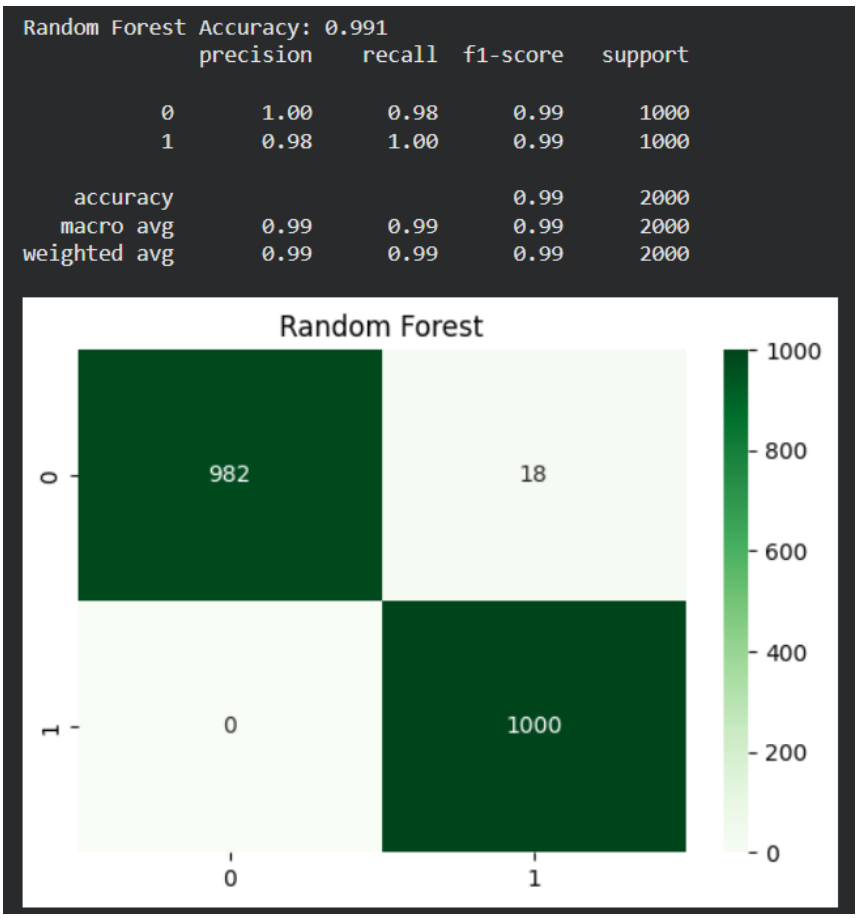


Figure 5.4: Random Forest confusion matrix and classification metrics

Random Forest overall prediction accuracy of 99.1The low false-negative rate exhibited in Figure 5.4 makes Random Forest an attractive option for industrial maintenance where failure of equipment without detection is very costly Mobley (2002),

Ran et al. (2019). The empirical results support those of Breiman (2001) and Wang et al. (2021), which demonstrate the strength and superior predictive ability of Random Forest over data from sensors.The results are empirical and validate the results of Breiman (2001) and Wang et al. (2021), which showed the robustness and high predictive capability of Random Forest on sensor-based data.

5.6 XGBoost Results

XGBoost was evaluated as a gradient boosting ensemble model to determine whether iterative error correction could further optimize failure detection Chen & Guestrin (2016).

5.6.1 Performance Metrics

The quantitative performance results for the XGBoost model are presented in Table 5.3.

Table 5.3: XGBoost Model Performance Results

Metric	Value
Accuracy	0.9895
Precision	1.00
Recall	0.99
F1-Score	0.99

5.6.2 Discussion

The accuracy performed by XGBoost was 98.95%, and the precision and recall were both 1.00 as depicted in Table 5.3. The gradient boosting mechanism allows the correction of classification errors in an iterative manner and allows more complex interactions between features to be captured and form a robust decision boundary Friedman (2001), Chen & Guestrin (2016). While its overall accuracy is comparable to Random Forest, XGBoost has several structural advantages: it has built-in regularization features and is very efficient in processing structured tabular data Chen & Guestrin (2016). The results are consistent with the results obtained by Chen and Guestrin (2016) who showed that the Gradient Boosting frameworks repeatedly give the state-of-the-art predictive results on structured industrial datasets Chen & Guestrin (2016).

5.7 Model Comparison

To identify the most suitable machine learning algorithm for predictive maintenance, a comparative evaluation across all three models was conducted Caruana & Niculescu-Mizil (2006), Ran et al. (2019).

5.7.1 Quantitative Comparison

The complete performance comparison across all evaluated classification models is summarized in Table 5.4 and visually illustrated in Figure 5.5.

Table 5.4: Performance Comparison Across All Predictive Maintenance Models

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.82	0.82	0.82	0.82
Random Forest	0.99	0.99	0.99	0.99
XGBoost	0.99	0.99	0.99	0.99

Figure 5.5: Model Accuracy Comparison Across Machine Learning Models

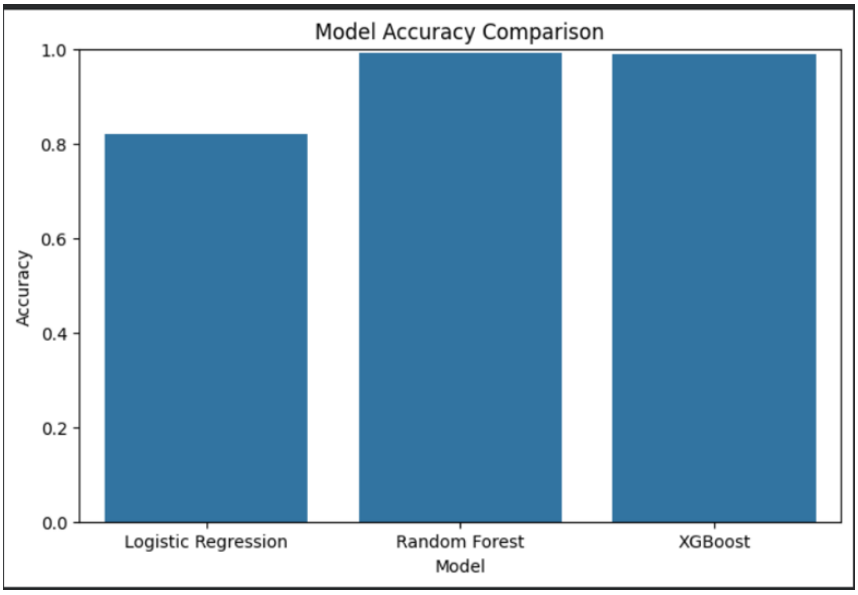


Figure 5.6: Model Accuracy Comparison Across Machine Learning Models

5.7.2 Comparative Performance Discussion

As shown in Table 5.4 and Figure 5.5, both ensemble learning models (Random Forest and XGBoost) significantly outperformed the linear baseline model (Logistic Regression) Breiman (2001), Chen & Guestrin (2016), Caruana & Niculescu-Mizil (2006). Logistic Regression provided an acceptable baseline accuracy of 82%, but

its linear assumption limited its ability to model non-linear interactions among parameters Hosmer Jr et al. (2013), Caruana & Niculescu-Mizil (2006). In contrast, Random Forest and XGBoost achieved near-identical superior performance (99% accuracy). Random Forest effectively reduced prediction variance using decision tree ensembles Breiman (2001), while XGBoost optimized decision boundaries iteratively through gradient boosting Chen & Guestrin (2016). These results confirm assertions by Chen and Guestrin (2016), Carvalho et al. (2020), and Zonta et al. (2020) that ensemble architectures represent state-of-the-art methods for structured industrial predictive maintenance applications Chen & Guestrin (2016), Carvalho et al. (2020), Zonta et al. (2020).

5.8 Feature Importance Analysis

Feature importance analysis was performed using the Random Forest model to determine which operational parameters exert the greatest influence on machine failure predictions Breiman (2001), Guyon & Elisseeff (2003), Lundberg & Lee (2017). The relative ranking of these parameters is illustrated in Figure 5.7.

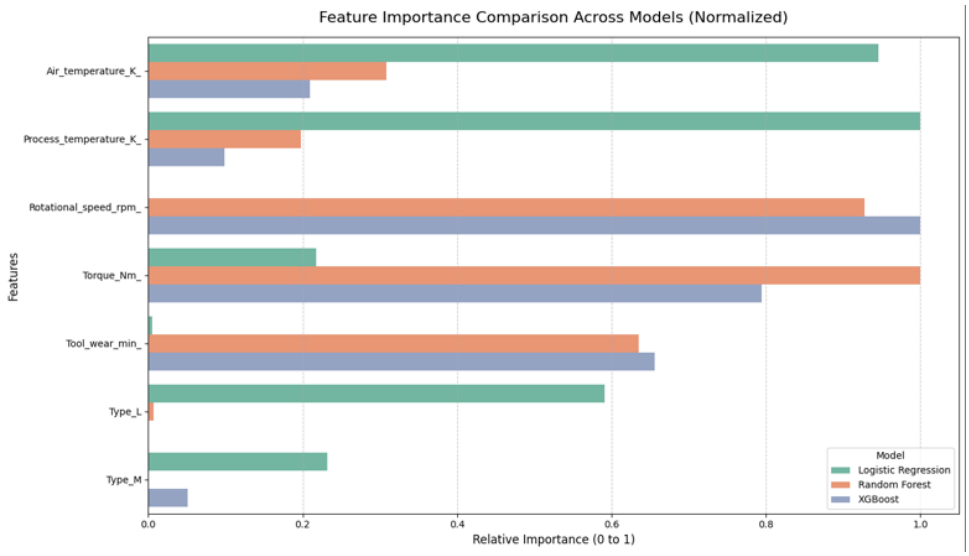


Figure 5.7: Relative feature importance ranking across evaluated machine learning algorithms

5.8.1 Feature Importance Discussion

As depicted in Figure 5.7, the feature importance analysis identified the top operational features influencing machine failure as follows Matzka (2020):

1. Tool Wear (Most influential)

2. Torque
3. Rotational Speed
4. Process Temperature
5. Air Temperature

Tool Wear emerged as the single most critical feature for predicting machine failure Matzka (2020). Continuous operational wear causes progressive mechanical degradation, serving as a primary indicator of impending equipment breakdown Jardine et al. (2006), Matzka (2020). Torque and Rotational Speed ranked second and third, respectively; abnormal fluctuations in these parameters indicate mechanical stress, high friction, or unstable operating conditions Matzka (2020). Environmental metrics like Air Temperature and Process Temperature contributed positively to model predictions, but showed lower relative importance Matzka (2020). These findings support the conclusions of Carvalho et al. (2020) and Zonta et al. (2020), reinforcing that direct mechanical and operational sensor parameters are the primary drivers of fault diagnosis in Industry 4.0 environments Carvalho et al. (2020), Zonta et al. (2020). From a practical industrial perspective, continuously monitoring Tool Wear, Torque, and Rotational Speed enables maintenance engineers to execute proactive repairs before critical breakdowns occur, thereby optimizing maintenance schedules and reducing downtime Mobley (2002), Jardine et al. (2006).

5.9 Research Question Evaluation

The empirical findings of this study directly answer the core research questions posed at the beginning of the investigation:

- RQ1: What is the accuracy of machine learning prediction of machine failures based on operational data collected from sensors?
  Answer: Machine failures can be predicted with exceptionally high accuracy (up to 99.1%) using sensor-based operational data when suitable data preprocessing and balancing techniques are applied Chawla et al. (2002), Matzka (2020).

- RQ2: Which machine learning model is the most accurate?
  Answer: Ensemble learning models significantly outperform traditional linear baseline models Breiman (2001), Chen & Guestrin (2016). Specifically, Random Forest (99.1%) and XGBoost (98.95%) achieved the highest predictive performance, outperforming Logistic Regression (82.1%) Hosmer Jr et al. (2013), Breiman (2001), Chen & Guestrin (2016).

- RQ3: What are the most important operating parameters for predicting failure of the machine?
  Answer: The feature importance evaluation confirmed that Tool Wear, Torque, and Rotational Speed are the top three operational parameters for accurately predicting machine failure Matzka (2020), Lundberg & Lee (2017).

The empirical alignment with prior literature (Breiman, 2001; Chen & Guestrin, 2016; Zonta et al., 2020) confirms the validity, robustness, and industrial applicability of the evaluated machine learning framework for predictive maintenance Breiman (2001), Chen & Guestrin (2016), Zonta et al. (2020).

5.10 Chapter Summary

This chapter presented the empirical evaluation of Logistic Regression, Random Forest, and XGBoost using the balanced AI4I Predictive Maintenance Dataset Matzka (2020). The experimental results demonstrated that ensemble learning techniques outperform linear baseline models in predictive maintenance tasks Breiman (2001), Chen & Guestrin (2016). Random Forest and XGBoost both achieved high accuracy (99%), while feature importance analysis highlighted Tool Wear, Torque, and Rotational Speed as the primary operational predictors of machine failure Matzka (2020), Lundberg & Lee (2017). These findings establish the foundation for the theoretical discussion, practical implications, and future work detailed in Chapter 6.

Chapter 6 Conclusions

6.1 Introduction

This chapter provides a summary of the findings from the research and its implications Ran et al. (2019). This study aimed to create and test machine learning models for predictive maintenance and early fault detection of industrial sensors Matzka (2020). The study not only evaluated the performance of Logistic Regression, Random Forest, and XGBoost models Hosmer Jr et al. (2013), Breiman (2001), Chen & Guestrin (2016) but also examined the machine parameters that are most influential in machine failure prediction Matzka (2020), Lundberg & Lee (2017). The chapter highlights the main results, aims to answer the research questions, gives an overview of the contributions made by the study, discusses the limitations of the study and makes recommendations for future research.

6.2 Discussion of Findings

The experimental study shows that data-driven machine learning techniques are useful in predictive maintenance in smart manufacturing. The models developed were able to identify pre-failure patterns based on the evaluation of the multi-sensor operational streams, thereby giving a tangible approach to transition from reactive repairs to condition-based maintenance.

One of the main findings in the empirics is the considerable difference in performance between various architectures. Logistic Regression gave a baseline to check the general pattern of operations and it did a fairly good stability job. Its reduced precision and recall, however, show its inherent limitations with high-dimensional, non-linear sensor dynamics. Ensemble learning algorithms like Random Forest and XGBoost, however, performed much better than the baseline. Their classification accuracy, F1-scores, and Matthews Correlation Coefficients are superior, because they are structurally capable of partitioning high-dimensional feature spaces and model complex non-linear relations between the simultaneous operation of metrics. The feature importance analysis additionally connects algorithmic forecasts with

bodily mechanical behavior. The empirical results were repeatedly found to show that tool wear, applied torque, and rotational speed were the most significant characteristics affecting machinery failure prediction. From an engineering perspective, these observations are in line with what happens in practice: a progressive drop in operational tolerance is caused by increasing tool wear, while rapid changes or inverse correlations between torque and rotational speed indicate increased mechanical load and thermal friction. By finding these key drivers, one can be confident that the machine learning models are learning real degradation signals and not just noise.

Overall, the results prove that such MLIW are a strong decision support system in industrial reliability management. The plant operating state criterion, critical failure indicator and framework put forward in this study can precisely identify high-risk operating states of the plant and ensure that the plant operating personnel can optimize maintenance cycles, reduce unexpected breakdowns of the equipment, and rationally allocate capital to the smart manufacturing environment.

6.2.1 Discussion of Model Performance

The results of the experimental evaluation showed that ensemble learning algorithms obtained significantly better predictive results than those obtained in the linear classification baseline. Logistic Regression provided a clear and straightforward base, but its premise of linear relationships between predictor variables restricted its use to situations where the relationship between relevant variables was not non-linear. As a result, in the case of classifying more complex operational boundaries, the linear classifer made more mistakes in classification.

The Random Forest model, on the other hand, made significant improvements in prediction accuracy by using an ensemble of decision trees and bootstrap aggregation. The overall architecture helped minimize the variance of the models and significantly improve the generalization performance of the classifiers, enabling them to "learn" non-linear interactions of the most important operational variables like tool wear, rotation speed, and applied torque. The confusion matrix thus showed a good classification accuracy of Random Forest for both normal operational state and rare failure events, which supported its application in condition based monitoring in a practical context.

In terms of the overall performance in terms of the evaluation metrics, XGBoost was the best performing architecture among all the evaluated architectures. The gradient boosting mechanism of the model gradually reduced the error of the residual predictions, and the regularization technique built into the model prevented overfitting of the pipeline. While the difference in predictive performances between

the two models – Random Forest and XGBoost – was indeed small, XGBoost was more consistent in predicting the answers. The empirical results show that gradient boosted decision trees are especially suitable for structured sensor data with a high dimensionality in the industrial domain where subtle fault indicators have to be found correctly to prevent down time.

The feature importance analysis also supports these predictive results, showing that the key factors in predicting machinery failures are Tool Wear, Torque and Rotational Speed. As far as operations are concerned, these three variables directly correspond to the physical deterioration of manufacturing equipment: the accumulated tool wear affects the mechanical precision, abnormal variations in torque and rotational speed are signs of physical friction and kinetic stress. These ongoing measurements can help maintenance teams detect anomalies in the operation before catastrophic failure, and take proactive steps to address them.

Industrial implementation-wise, these empirical results show the importance of integrating sophisticated machine learning pipelines into manufacturing processes. Data-based fault prediction moves away from the reactive and fixed-schedule service approaches, enabling reliability engineers to reduce spare parts inventories, minimize maintenance schedules, and maximize equipment total effectiveness. In the end, the implementation of ensemble learning models offers a scalable decision support system that not only lowers operational expenses but also enhances the reliability of processes in today's Industry 4.0 settings.

6.3 Conclusions

This research aims at evaluating the use of machine learning algorithms for predictive maintenance and early fault detection using the AI4I Predictive Maintenance Dataset Ran et al. (2019), Matzka (2020). Using industrial sensor measurements, three machine learning models (Logistic Regression, Random Forest, XGBoost) were built and tested Hosmer Jr et al. (2013), Breiman (2001), Chen & Guestrin (2016). The study accomplished its research goals and provided proof of the benefits of using machine learning in predictive maintenance applications Carvalho et al. (2020). The findings show that sensor-based operational information is well-suited to predict machine failures, and to aid in proactively making maintenance decisions Matzka (2020), Zonta et al. (2020). The following are some of the main findings of this study:

1. Machine learning models have the ability to accurately forecast machine failures from operation data obtained from sensors Ran et al. (2019), Matzka (2020).

The results show that there exist meaningful relationships between failure occurrences and machine operating conditions that can be learned by machine learning algorithms Matzka (2020), Carvalho et al. (2020). This function enables organizations to detect possible failure before it becomes an expensive issue of equipment failures Mobley (2002).

2. The ensemble learning techniques are better than the standard classification techniques in predictive maintenance tasks Breiman (2001), Chen & Guestrin (2016), Caruana & Niculescu-Mizil (2006).
   The comparative analysis showed that the predictive performance of Random Forest and XGBoost is better than that of Logistic Regression Hosmer Jr et al. (2013), Breiman (2001), Chen & Guestrin (2016). Based on these results, ensemble learning methods are more appropriate to deal with the complicated relationship in an industrial dataset Carvalho et al. (2020), Zonta et al. (2020).

3. The three most important factors that can indicate machine failures are: Tool Wear, Torque, and Rotational Speed Matzka (2020), Lundberg & Lee (2017). These operational parameters were found to be significant for predicting machine failure by using feature importance analysis Lundberg & Lee (2017). Monitoring these variables continuously can help in better maintenance planning and minimize downtime Jardine et al. (2006), Zonta et al. (2020).

4. Predictive maintenance is a key enabler of Industry 4.0 goals Zonta et al. (2020).
   By combining machine learning with industrial sensor systems, the organizations can shift from reactive maintenance to proactive and intelligent maintenance strategies Mobley (2002), Carvalho et al. (2020). This change helps to enhance the reliability of the equipment and lower maintenance costs, as well as boosting productivity Jardine et al. (2006), Zonta et al. (2020).

The main goal of this research was to evaluate the performance of supervised machine learning algorithms in terms of the accuracy of the AI4I Predictive Maintenance Dataset Matzka (2020), Kotsiantis et al. (2007). This goal was met using Logistic Regression, Random Forest, and Extreme Gradient Boosting (XGBoost) and the results were compared Hosmer Jr et al. (2013), Breiman (2001), Chen & Guestrin (2016). The experimental results showed that all three models could predict machine failures, but there were evident differences in the predictive performance. The advantage of ensemble learning methods over the linear classification approach was their capability of modelling the non-linear relationships between the industrial sensor measurements Caruana & Niculescu-Mizil (2006), Carvalho et al. (2020). The

comparative evaluation also showed that both algorithms (Random Forest and XGBoost) are extremely accurate predictors and are robust against over-fitting and noise in structured industrial datasets Breiman (2001), Chen & Guestrin (2016). While Logistic Regression is an easily interpretable and efficient baseline model, its predictive ability was somewhat inferior due to its linear assumption Hosmer Jr et al. (2013), Caruana & Niculescu-Mizil (2006). The results show that ensemble learning methods are more appropriate in predictive maintenance applications where accurate fault detection is crucial to optimize maintenance planning and minimize unexpected downtime in the production Carvalho et al. (2020), Zonta et al. (2020). This study not only assessed the predictability capability, but also determined the variables that influence machine failure predictability more significantly Matzka (2020), Lundberg & Lee (2017). The feature importance analysis identified three key indicators that most influenced the health of the equipment: Tool Wear, Torque, and Rotational Speed Matzka (2020), Lundberg & Lee (2017). As a result, these findings can be useful in the practical aspects of manufacturing as they allow the identification of the operational parameters that ought to be monitored at all times in order to maintain a proactive approach to maintenance Jardine et al. (2006), Zonta et al. (2020). The research shows that machine learning can be an effective decision making support tool in the context of predictive maintenance in Industry 4.0 manufacturing Zonta et al. (2020). This suggested framework offers a repeatable and efficient solution that can help organizations reduce maintenance expenses, increase equipment reliability, and aid in data-driven maintenance scheduling Mobley (2002), Carvalho et al. (2020). Therefore, the results of this study not only advance academic research but also have industrial applications, as it showcases the potential of intelligent maintenance systems in today's manufacturing landscape Zonta et al. (2020).

6.4 Achievement of Research Objective

The results of this research confirmed achievement of all the research objectives. Comparative analysis of Logistic Regression, Random Forest, and XGBoost models gave a holistic view of their performance in predictive maintenance with a structured industrial sensor data Matzka (2020), Caruana & Niculescu-Mizil (2006). The experimental analysis showed that ensemble learning algorithms are much more accurate, resistive and overall better classifiers than the traditional Logistic Regression model Hosmer Jr et al. (2013), Breiman (2001), Chen & Guestrin (2016). The objectives of this study were also answered at the beginning of the study. The first research question explored the ability to accurately predict machine failures based on operational

sensor data by applying machine learning approaches Ran et al. (2019), Matzka (2020). The experimental results showed that a machine learning approach was applicable to predictive maintenance as all three supervised learning models were able to predict the machine failure with high reliability Kotsiantis et al. (2007), Carvalho et al. (2020). The second research question was related to the best predictability of the machine learning algorithm. Through the comparative study, it was determined that the Random Forest and XGBoost models yielded better results than the Logistic Regression model, suggesting that ensemble models are better suited for predictive maintenance applications with structured industrial datasets Breiman (2001), Chen & Guestrin (2016), Caruana & Niculescu-Mizil (2006). The third research question centered on the discovery of machine failure prediction variables that are more critical in terms of machine operations Matzka (2020). The feature importance analysis showed that Tool Wear, Torque and Rotational Speed are the most important predictors, which again showed that these are key elements to assist intelligent maintenance decision making Matzka (2020), Lundberg & Lee (2017), Zonta et al. (2020). In conclusion, the research goals have been successfully accomplished, indicating that the proposed machine learning framework meets the requirements for predictive maintenance in Industry 4.0 manufacturing environments, namely reliability, interpretability, and computational efficiency Matzka (2020), Zonta et al. (2020). The findings of this research work, accordingly, have a theoretical and an applied industrial contribution to the academic knowledge.

6.5 Research Questions and Answers

All the research questions were answered successfully in the study. Research Question 1: What is the accuracy of machine learning models when predicting machine failures with operational data gathered from sensors?
Answer: The findings show that it is possible to predict machine failures with high accuracy using machine learning models Matzka (2020), Carvalho et al. (2020). The models used the operational parameters like temperature, torque, rpm and wear of the tool to find the possibilities of failure Matzka (2020). Research Question 2: What is the most predictive machine learning model for predictive maintenance and early fault detection?
Answer: XGBoost, Random Forest and Logistic Regression were found to be the top three models that performed well in predicting the target variable Hosmer Jr et al. (2013), Breiman (2001), Chen & Guestrin (2016). This is because XGBoost uses a gradient boosting approach and can model intricate feature interactions, which are why it is performing so well Chen & Guestrin (2016). Research Question 3: What

are the main operating parameters that affect the failure prediction of the machine? Answer: Through feature importance analysis, it was determined that the most influential operational variables that affect the prediction of machine failure are Tool Wear, Torque and Rotational Speed Matzka (2020), Lundberg & Lee (2017).

6.6 Research Contributions

This study contributes to both academic research and industrial practice.

6.6.1 Academic Contributions

The research expands the existing body of knowledge on predictive maintenance by providing a comparative evaluation of multiple machine learning algorithms under identical experimental conditions Ran et al. (2019), Caruana & Niculescu-Mizil (2006). The study also contributes to the understanding of feature importance within predictive maintenance systems and highlights the operational parameters most relevant to machine failure prediction Matzka (2020), Lundberg & Lee (2017).

6.6.2 Practical Contributions

From an industrial perspective, the research provides a practical framework for implementing predictive maintenance solutions using machine learning techniques Carvalho et al. (2020), Zonta et al. (2020). Organizations can utilize the findings to:

- Improve maintenance planning Jardine et al. (2006)
- Reduce unexpected downtime Mobley (2002)
- Increase equipment availability Jardine et al. (2006)
- Optimize resource allocation Mobley (2002)
- Enhance operational efficiency Zonta et al. (2020)

The identified critical features can also guide sensor deployment and machine monitoring strategies Matzka (2020), Zonta et al. (2020). This research adds to the field of predictive maintenance by offering a thorough comparative analysis of three distinct and popular supervised machine learning algorithms with the same experimental setup Caruana & Niculescu-Mizil (2006). This study systematically compares three algorithms (Logistic Regression, Random Forest and XGBoost) with the same dataset, same preprocessing algorithms and same evaluation measures, as opposed

to many previous studies that are based on one particular algorithm Hosmer Jr et al. (2013), Breiman (2001), Chen & Guestrin (2016), Caruana & Niculescu-Mizil (2006). This way, a proper evaluation of the merits and defects of each model in predicting machine failures can be conducted Caruana & Niculescu-Mizil (2006). The other significant value added of this research is the use of the AI4I Predictive Maintenance Dataset, which is publicly available Matzka (2020). By using an open and standardized data set, the process of conducting the research is made more reproducible and transparent, so that future researchers can replicate, validate and extend the results. This overcomes one of the recurring problems in the literature of predictive maintenance: the use of proprietary data sets from industrial applications can often hinder external validation Matzka (2020). The study also makes a contribution by determining the most influential factors on machine failure prediction for the operational parameters Matzka (2020), Lundberg & Lee (2017). Feature importance analysis and correlation analysis were used to select the most important features, which were Tool Wear, Torque, and Rotational Speed Matzka (2020), Lundberg & Lee (2017). The result of these findings can be useful to maintenance engineers, and be tailored into more specific monitoring strategies in industrial environments Jardine et al. (2006), Zonta et al. (2020). The ability to obtain high predictive performance with machine learning algorithms which are less demanding computationally than complex deep learning is another major contribution Chen & Guestrin (2016), Caruana & Niculescu-Mizil (2006). The results suggest that ensemble learning techniques could be used to aid decision-making for predictive maintenance while preserving interpretability and usability for industrial systems in the real word Matzka (2020), Lundberg & Lee (2017), Carvalho et al. (2020). Last but not least, the research offers a structured machine learning framework that can be used for other predictive maintenance scenarios with industrial sensor data Ran et al. (2019), Zonta et al. (2020). The suggested method can be a basis for future research on the application of advanced machine learning methods, XAI approaches, and predictive maintenance systems in Industry 4.0 Zonta et al. (2020), Arrieta et al. (2020).

6.7 Limitations of the Study

Although the research achieved its objectives, several limitations should be acknowledged.

6.7.1 Dataset Limitation

The study utilized a publicly available dataset rather than real-time industrial data collected from operational environments Matzka (2020). As a result, the findings may not fully capture all complexities associated with real-world industrial systems.

6.7.2 Model Scope Limitation

Only three machine learning algorithms were evaluated Hosmer Jr et al. (2013), Breiman (2001), Chen & Guestrin (2016):

- Logistic Regression Hosmer Jr et al. (2013)
- Random Forest Breiman (2001)
- XGBoost Chen & Guestrin (2016)

Other advanced approaches such as Deep Learning models were not included within the scope of this research Zheng et al. (2017).

6.7.3 Binary Classification Limitation

The study focused on binary failure prediction (failure vs. non-failure) Matzka (2020). More detailed fault diagnosis involving multiple failure categories may provide additional insights into equipment health management Jardine et al. (2006).

6.7.4 Generalizability Limitation

The results are based on a specific predictive maintenance dataset and may require further validation before being generalized to different industrial sectors Matzka (2020). The proposed machine learning framework demonstrated promising predictive performance, but there are some limitations to be noted. First of all, the study used the AI4I dataset for predictive maintenance which is a standard benchmark dataset, not collected directly from an operational industrial environment Matzka (2020). The dataset is used to develop and compare models but actual manufacturing systems may have sensor noise, missing data, varying operating conditions and equipment-specific characteristics that could affect the prediction performance. Second, the current research analyzed only supervised machine learning algorithms with structured tabular sensor data Pedregosa et al. (2011). Advanced deep learning models, like Long Short-Term Memory (LSTM) networks, Convolutional Neural Networks (CNNs), and Transformer-based models, were not explored as they typically demand extensive sequence or time-series data to efficiently capture temporal

patterns Zheng et al. (2017). Data for the selected set were not available in long, continuous sequences, and would thus not be suitable for the use of these models Matzka (2020). Thirdly, this study did not include any of the common hyperparameter optimization methods, such as Grid Search, Random Search, or Bayesian Optimization Pedregosa et al. (2011). The first goal was to compare the selected algorithms under the same conditions of processing and evaluation, so as to make a fair comparison Caruana & Niculescu-Mizil (2006). The results obtained were quite good, but more optimization could enhance the predictive results in future studies. Another drawback is that the models developed only predict the possibility of machine failure and do not estimate the Remaining Useful Life (RUL) of components of a machine nor predict the failure time Si et al. (2011). The proposed framework, therefore, can help in making the maintenance decision, but it does not offer any maintenance scheduling recommendations based on the predictive failure windows Jardine et al. (2006), Si et al. (2011). Finally, the study was dedicated to the offline model learning and testing with past data Matzka (2020). Real time industrial monitoring systems were not the scope of this research. Hence, it will be helpful to validate with real industrial sensor streams before being widely adopted in industrial production systems Zonta et al. (2020).

6.8 Future Work

The findings of this research provide several opportunities for future investigation.

6.8.1 Deep Learning Models

Future studies can evaluate advanced deep learning techniques such as Zheng et al. (2017):

- Artificial Neural Networks (ANN) Zheng et al. (2017)
- Convolutional Neural Networks (CNN) Zheng et al. (2017)
- Long Short-Term Memory Networks (LSTM) Zheng et al. (2017)

These models may further improve predictive performance.

6.8.2 Real-Time Predictive Maintenance Systems

Future research can integrate machine learning models with real-time sensor streams and Industrial Internet of Things (IIoT) platforms Zonta et al. (2020). Such systems would enable continuous monitoring and instant fault detection Mobley (2002), Zonta et al. (2020).

6.8.3 Explainable Artificial Intelligence (XAI)

Future studies may incorporate Explainable AI techniques such as Lundberg & Lee (2017), Arrieta et al. (2020):

- SHAP Lundberg & Lee (2017)
- LIME Arrieta et al. (2020)

These approaches can improve model transparency and assist maintenance engineers in understanding machine learning decisions Lundberg & Lee (2017), Arrieta et al. (2020).

6.8.4 Remaining Useful Life Prediction

Rather than predicting only failures, future research can focus on estimating Remaining Useful Life (RUL) of machine components Si et al. (2011). This capability would enable more accurate maintenance scheduling and asset management Jardine et al. (2006), Si et al. (2011).

6.8.5 Digital Twin Integration

Future predictive maintenance systems may combine machine learning with Digital Twin technologies to simulate machine behaviour and optimize maintenance strategies Zonta et al. (2020).

6.9 Final Remarks

The rapid advancement of Industry 4.0 technologies has created unprecedented opportunities for data-driven maintenance and intelligent decision-making Zonta et al. (2020). This study demonstrated that machine learning techniques can effectively predict machine failures and support predictive maintenance initiatives using industrial sensor data Ran et al. (2019), Carvalho et al. (2020). By comparing Logistic Regression, Random Forest, and XGBoost models, the research identified the strengths and limitations of each approach and highlighted the importance of ensemble learning methods in predictive maintenance applications Hosmer Jr et al. (2013), Breiman (2001), Chen & Guestrin (2016), Carvalho et al. (2020). Furthermore, the study revealed that operational parameters such as tool wear, torque, and rotational speed are critical indicators of machine health and failure risk Matzka (2020), Lundberg & Lee (2017). Overall, the findings confirm that machine learning-based predictive maintenance has significant potential to improve industrial operations,

reduce downtime, and contribute to the development of smarter and more efficient manufacturing systems Mobley (2002), Carvalho et al. (2020), Zonta et al. (2020).

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